PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

DESIGN AND DEVELOPMENT OF A DIGITAL SALES PROCESSING SYSTEM WITH CENTRALIZED SALES MANAGEMENT AND ANALYTICAL REPORTING FOR LUNA'S ARROZCALDO — ILONGGO LEGACY SINCE 1967

A Capstone Project

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ABSTRACT

The Digital Sales Processing System developed in this study is a web-based platform developed specifically for Luna's Arrozcaldo, a beloved Ilonggo food institution that has been serving the community of Iloilo City since 1967. The system integrates a digital Point-of-Sale (POS) terminal, centralized sales management, real-time inventory tracking, customer relationship management, promotional campaign tools, and analytical reporting in a unified platform built on PHP and PostgreSQL (Supabase).

Luna's Arrozcaldo currently operates across eight branches in Iloilo City using entirely manual, paper-based processes for sales recording, inventory monitoring, and performance tracking. This reliance on paper has created compounding operational problems: arithmetic errors in daily sales summaries, the owner's need to physically travel between branches each day to collect performance data, inability to detect stockouts before service is disrupted, and the absence of any mechanism for identifying and rewarding loyal customers.

The proposed system addresses these problems by providing a role-based, multi-branch digital platform that allows transactions to be processed and recorded digitally at each branch POS terminal, with all data aggregating automatically into a centralized dashboard accessible to the business owner from any location. The system's analytical reporting module transforms accumulated transaction data into actionable business intelligence through interactive charts, branch performance comparisons, and menu item rankings.

The system was developed using the Agile methodology, with iterative development cycles guided by continuous feedback from the client organization. User acceptance testing was conducted with representative cashier, branch manager, and administrator users, confirming the system's usability, accuracy, and practical value. The technology stack comprises PHP for the backend, PostgreSQL hosted on Supabase for the database, HTML/CSS/JavaScript for the frontend, and Docker for containerized deployment.

The findings of this study demonstrate that a carefully designed, appropriately scoped digital system can meaningfully transform the operations of a heritage food MSME, improving efficiency, accuracy, and management visibility while preserving the warmth and community spirit that define the institution's identity.

Keywords: Point of Sale, Digital Sales Processing, Centralized Management, Analytical Reporting, MSME, Philippine Food Heritage, Iloilo, Agile Development, PostgreSQL, PHP

TABLE OF CONTENTS

LIST OF FIGURES

Figures

CHAPTER I

INTRODUCTION

This chapter presents the foundational elements of the capstone research entitled Design and Development of a Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo. It begins by establishing the historical and cultural context of Luna's Arrozcaldo, an Ilonggo food institution that has been serving the community of Iloilo City since 1967. The chapter then articulates the specific operational problems that motivated this research, defines the study's objectives and boundaries, and establishes the significance of the work for multiple stakeholder groups. Theoretical and conceptual frameworks ground the research in established academic traditions, and the chapter closes with a full profile of the research locale and the client organization, as well as a declaration of the ethical commitments that have guided every phase of this study.

The researchers wish to acknowledge from the outset that this chapter — and this entire capstone project — is animated by a genuine sense of responsibility toward a business and a community that have given so much to Iloilo City. Luna's Arrozcaldo is not merely a case study. It is a 57-year-old institution whose continued health and growth matter to thousands of Ilonggos who have grown up eating its food, celebrating milestones at its tables, and finding comfort in its familiar flavors during difficult times. The digital system proposed in this study is offered in that spirit: as a tool in service of a legacy, not a replacement for it.

Background of the Study

Agriculture and food culture are deeply embedded in the identity of Iloilo City. Among the many culinary institutions that have shaped the taste and tradition of this Western Visayas capital, Luna's Arrozcaldo stands in a category of its own. Founded in 1967 in the La Paz district by the woman her community knew as Lola Luna, the establishment began as a modest carinderia offering the kind of honest, nourishing food that busy families, hardworking laborers, and hungry students needed at a price they could actually afford. At its heart was arrozcaldo — a thick, golden, ginger-perfumed rice porridge that tasted, to everyone who ate it, like the best version of something they had been eating their whole lives.

Lola Luna's arrozcaldo was not simply a recipe. It was a philosophy made edible: the conviction that good food should be accessible to everyone, that the act of feeding another person is an act of care, and that a business built on those principles would earn not just customers but community. This philosophy has proven prophetic. Over 57 years, Luna's Arrozcaldo has grown from a single carinderia to an eight-branch enterprise with locations spanning the major commercial and residential districts of Iloilo City — from Festive Walk Mall and SM City Iloilo to General Luna Street, Jaro, Molo, La Paz, Calumpang, and Tagbak. Each branch carries the same menu, the same quality standards, and the same spirit of generous community service that Lola Luna established at the very beginning.

The arrozcaldo that made Luna's famous is a dish with layered cultural history. Its name derives from the Spanish words arroz (rice) and caldo (broth), a linguistic inheritance from over three centuries of Spanish colonial rule in the Philippines. But the dish as it is prepared throughout the archipelago is unmistakably Filipino in character: comforting, generous, adaptable, and deeply nourishing. In Iloilo, arrozcaldo carries particular cultural resonance. The Ilonggos — speakers of Hiligaynon, one of the Philippines' most melodic regional languages — have long been recognized for a culinary tradition that is refined, bold, and rooted in a sophisticated understanding of flavor balance. Luna's version brings this Ilonggo sensibility to its fullest expression: native chicken from the highlands of Iloilo Province, locally sourced ginger prized for its aromatic intensity, rice cooked low and slow to release its starch into a silky, substantial broth, seasoned with fish sauce and finished with crispy fried garlic, bright green onions, and a perfectly cooked egg.

The business passed from Lola Luna to her children and then to her grandchildren, each generation honoring the original with deep respect while making thoughtful adaptations to serve evolving tastes and expanding opportunities. The second generation oversaw the first branch openings. The third generation, which manages the business today, has continued this expansion while beginning to grapple seriously with the operational challenges that come with running a multi-branch food enterprise in an increasingly competitive, increasingly digital marketplace.

Today, Luna's Arrozcaldo is recognized throughout Iloilo City and beyond as one of the defining culinary institutions of the Western Visayas region. Its name is synonymous with authentic Ilonggo comfort food. It is featured in local food guides and tourism materials. Visiting dignitaries, balikbayan returnees, and food tourists make it a point to eat at Luna's when they are in Iloilo. The brand carries a weight of cultural meaning and community affection that is, in the truest sense, priceless — and that the current generation of the Luna family carries with a deep sense of custodial responsibility.

Against this backdrop of extraordinary cultural richness, Luna's Arrozcaldo faces a set of operational challenges that are both urgent and entirely solvable. The business currently operates without any digital system for recording or managing sales, inventory, or customer information across its eight branches. All transactions are processed manually, all records are maintained on paper, and all performance monitoring requires the physical presence of the owner or designated managers at each branch location. The Digital Competitiveness Index data for the Philippines has consistently shown that while large enterprises have made significant progress in digital adoption, micro, small, and medium enterprises (MSMEs) — particularly in the food service sector — have lagged considerably behind.

The COVID-19 pandemic, devastating as it was for the food service industry, paradoxically accelerated digital awareness among many MSME owners. The experience of managing a business during lockdowns — when physical presence was impossible and remote communication was essential — made clear to many entrepreneurs the value of digital tools for monitoring and managing operations at a distance. Luna's Arrozcaldo's owners have expressed, in their interviews with the research team, a genuine readiness to embrace digital transformation, provided that any system introduced is practical, affordable, and appropriately tailored to their specific operational context. It is in direct response to this readiness that this capstone research was conceived and the proposed system developed.

The digital landscape in the Philippines has shifted dramatically in recent years. The proliferation of smartphones and mobile data networks have made digital tools accessible to businesses that could not have considered them a generation ago. For a business like Luna’s Arrozcaldo, the question is no longer whether digital tools are appropriate, but which tools are right and how they should be introduced in a way that honors rather than disrupts the institution's deeply embedded operational culture.

Luna's Arrozcaldo's most experienced staff carry an irreplaceable body of operational knowledge: which ingredients are needed in what quantities on which days, which branches run out of popular items first, which time periods generate the highest volumes. When this knowledge is not captured in any formal system, it exists only in individual memory — and when those staff members leave, the knowledge leaves with them. Formalizing this knowledge in a digital system is not merely a matter of efficiency; it is an act of organizational resilience.

These contextual realities — rising consumer expectations, an increasingly competitive food service landscape, vulnerable institutional knowledge, and growing accessibility of appropriate digital tools — together create both the imperative and the opportunity for Luna's Arrozcaldo's digital transformation.

Statement of the Problem

The fundamental problem facing Luna's Arrozcaldo is the absence of a unified, digital platform for managing its sales operations, inventory, and business performance data across its eight branch locations in Iloilo City. This absence creates a cascade of interconnected operational challenges that affect efficiency, accuracy, management oversight, and the long-term sustainability of the business. Each of these challenges is documented in the subsections that follow, drawing directly on information gathered through structured interviews with the business owner, branch managers, and cashier staff conducted as part of the preliminary research for this study.

It is important to note that the problems described here are not the result of negligence or poor management. They are the natural consequence of a business that grew organically, guided by family wisdom and community relationships rather than by formal business management systems, and that has simply not yet had access to the right digital tools to support its current scale of operations. The proposed system is designed to address these problems not by imposing an alien operational logic on a well-functioning organization, but by providing digital support for the workflows and management approaches that have already proven effective over five and a half decades.

Problem 1: The Manual Travel Burden

The business owner of Luna's Arrozcaldo currently devotes a significant portion of each workday to traveling between branch locations. This travel is necessary because, in the absence of any digital reporting system, the only way for the owner to obtain current, accurate information about each branch's performance is to visit in person. During these visits, the owner reviews handwritten sales logs, inspects cash drawer reconciliations, checks inventory levels, and meets briefly with branch managers to discuss any issues that have arisen.

On a typical day, this circuit of branch visits consumes between two and four hours — time that the owner characterizes, in interviews with the research team, as frustrating and exhausting rather than productive. Traffic conditions in Iloilo City, which has grown significantly in recent years and experiences significant congestion during morning and afternoon peak hours, frequently extend travel times beyond initial estimates. The opportunity cost of this travel is substantial: every hour spent driving between branches is an hour not spent on menu development, supplier relationship management, community engagement, or strategic planning.

Problem 2: Inefficient Paper-Based Record Keeping

At each Luna's Arrozcaldo branch, every sales transaction is recorded manually by the cashier on duty using a pre-printed order slip and a handwritten sales log. At the end of each shift, the cashier tallies all transactions by hand and prepares a shift summary, which is then reviewed and signed by the branch manager. These paper records are collected — either during the owner's daily visit or at the end of the week — and brought to a central location for further review and filing.

This process is slow, labor-intensive, and prone to error at every stage. Arithmetic mistakes in individual transaction totals compound when tallied into shift summaries. Illegible handwriting causes ambiguity when records are reviewed. Missing or damaged receipts create gaps in the audit trail. The research team's analysis of one month's paper records from a single branch identified discrepancies in more than fifteen percent of daily sales summaries — a rate of error that, projected across the entire eight-branch network and over an entire year, represents a significant source of financial uncertainty. Paper records are also inherently fragile; several branch managers recalled specific instances of water damage to stored records during the rainy season, resulting in the permanent loss of historical sales data.

<u>Problem 3: Absence of Real-Time Centralized Oversight</u>

The third major problem confronting Luna's Arrozcaldo is the absence of any mechanism for real-time, centralized monitoring of business performance across all eight branches. At any given moment during a business day, the owner and management team have no way of knowing — without physically visiting each location — how sales are progressing, what inventory levels look like, whether any operational issues have arisen, or how the current day is tracking against historical performance benchmarks.

This informational blindness has real operational consequences. A branch experiencing unexpectedly high customer volume cannot signal its need for additional staff support through any formal channel. A branch that has run out of a key ingredient cannot trigger an automatic reorder. A cashier making systematic errors cannot be detected through any monitoring mechanism until the owner's next physical visit. The cumulative effect of this oversight gap is a management environment characterized by reactive problem-solving rather than proactive planning.

Problem 4: Inventory and Food Waste Management Gaps

Effective inventory management is one of the most critical aspects of running a food service business. Food costs typically represent 25 to 35 percent of revenue in the food service industry, and the difference between a profitable and an unprofitable food business often comes down to how effectively food costs are managed. Luna's Arrozcaldo currently manages its inventory through a combination of experienced intuition and manual stocktaking conducted at irregular intervals. This approach, while benefiting from genuine staff expertise, is inherently imprecise. The result is a pattern of alternating problems: sometimes branches run short of key ingredients before the end of service; at other times, branches over-order, leading to spoilage of perishable ingredients and unnecessary cost increases.

Problem 5: Customer Retention and Loyalty Challenges

Luna's Arrozcaldo has built its customer base through the quality of its food and the warmth of its service rather than through formal customer retention programs. While this approach has been remarkably successful over 57 years, it leaves the business without the ability to systematically identify, reward, and deepen relationships with its most loyal customers. In an increasingly competitive Iloilo food service market — the city's restaurant scene has expanded dramatically in recent years — the ability to build and maintain customer loyalty through data-driven engagement is becoming an important competitive advantage. Currently, the business has no way of identifying which customers visit most frequently, which menu items are most popular among repeat customers, or which customers have not visited in an unusually long time.

Problem 6: Staff Training and Knowledge Transfer Gaps

The operational knowledge required to run a Luna's Arrozcaldo branch effectively is substantial. Cashiers must memorize menu items, prices, discount combinations, and the procedures for handling partial orders, customer complaints, or payment discrepancies. In the absence of formal digital documentation, all training is conducted through direct staff-to-staff transmission. New staff members have no reference document to consult when they encounter unfamiliar situations during busy shifts. The proposed digital system addresses this by creating a formal operational procedure through the POS interface itself, guiding users through the correct process for each transaction type and creating a consistent standard across all branches.

Problem 7: Absence of Sales Forecasting Capacity

Luna's Arrozcaldo currently has no systematic mechanism for demand forecasting beyond the general seasonal intuitions of experienced staff. The absence of historical sales data in any accessible form makes it impossible to identify patterns — day-of-week variations, seasonal peaks, the impact of local events on nearby branch traffic — that would make meaningful forecasting possible. The analytical reporting module of the proposed system generates the historical data archive from which forecasting insights can eventually be drawn, transforming the business from reactive to proactive in its operational planning.

Objectives of the Study

The objectives of this study are grounded in the specific, documented operational challenges described in the preceding section, and are designed to be concrete, measurable, and achievable within the scope of the capstone project. They reflect the researchers' commitment to delivering not a theoretical demonstration of technical capability but a genuinely useful system that creates real, lasting value for Luna's Arrozcaldo and its stakeholders.

General Objective

The primary objective of this study is to design and develop a Digital Sales Processing System with Centralized Sales Management and Analytical Reporting specifically tailored to the operational context, cultural identity, and business goals of Luna's Arrozcaldo. The system is intended to modernize the business's sales operations, enhance its capacity for data-driven management, and strengthen its ability to deliver consistent, high-quality service across all eight branch locations — while honoring and preserving the warmth, authenticity, and community spirit that have defined Luna's for more than half a century.

Specific Objectives

In pursuit of this general objective, this study specifically aims to accomplish the following:

1. To eliminate the need for the business owner to travel daily between branches for the purpose of collecting sales data and monitoring operations, by providing a secure, web-based management dashboard through which the owner can view real-time performance data from all branches simultaneously, from any location with an internet connection, at any time of day.

2. To replace the current paper-based sales recording system with a digital Point-of-Sale terminal that enables cashiers to record transactions quickly, accurately, and consistently. The POS system will automatically compute transaction totals, apply applicable discounts and promotions, generate digital receipts, and update the central sales database in real time, eliminating the arithmetic errors, illegibility issues, and transcription mistakes that characterize the current manual system.
3. To implement a centralized sales management platform that aggregates operational data from all eight Luna's Arrozcaldo branches into a single, unified PostgreSQL database hosted on Supabase. This platform will serve as the authoritative single source of truth for all sales, inventory, customer, and staff data across the entire branch network.
4. To develop an analytical reporting module that transforms raw sales data into actionable business intelligence through the generation of daily, weekly, and monthly sales reports; branch performance comparisons; menu item rankings; and trend analyses, presented in visual formats — charts, graphs, and dashboards — accessible and interpretable by users without specialized data analysis training.
5. To implement a real-time inventory management module that tracks stock levels automatically as sales are processed through the POS terminal, generates alerts when items fall below configurable threshold levels, and maintains a complete, auditable digital record of all inventory movements including deliveries, adjustments, and inter-branch transfers.
6. To develop a customer management feature that enables the creation and maintenance of digital customer profiles linked to purchase histories, supporting the

identification of loyal customers and the design of targeted promotional campaigns, in full compliance with the Data Privacy Act of 2012 (Republic Act 10173).

7. To design and implement a coupon and promotions management module enabling management to create digital discount offers, define their terms and validity periods, deploy them to specific branches or the entire network, and track their usage and effectiveness through the analytical reporting system.
8. To deliver a role-based access control system ensuring that each user — whether a cashier, branch manager, or business owner/administrator — can access only the features and data appropriate to their role, protecting sensitive business information and ensuring compliance with data governance best practices.
9. To ensure that the proposed system is genuinely usable by the actual people who will operate it — cashiers, branch managers, and the business owner — through a user interface design process that prioritizes clarity, simplicity, and appropriateness to the food service work environment, validated through formal usability testing with representative users before final delivery.
10. To document the design, development, and evaluation of the proposed system in a form that is academically rigorous, practically useful as an operational guide for Luna's Arrozcaldo's management team, and accessible as a reference for future researchers working on similar problems in the Philippine MSME context.

Significance of the Study

The significance of this study must be understood on multiple levels — operational, cultural, economic, educational, and social. The researchers have made a deliberate effort throughout this project to think beyond the narrow question of whether the proposed system will work technically, and to ask the broader question of whether it will matter — to real people, in a real community, over a meaningful period of time.

For Luna's Arrozcaldo and the Luna Family

The most immediate and direct beneficiaries of this research are the members of the Luna family who own and operate Luna's Arrozcaldo, and the staff members who bring the business to life every day. For the business owner, the proposed system represents a fundamental transformation in the experience of managing the enterprise. The daily grind of branch-to-branch travel, the anxiety of not knowing how the business is performing in real time, the frustration of reconciling paper records that never quite add up — all of these are addressable through the proposed system, and their resolution will free the owner to invest time and energy in higher-value activities such as menu development, supplier partnerships, and community engagement.

For branch managers, the proposed system provides something that has been largely absent from their professional experience: a clear, comprehensive, and current picture of their branch's performance at any given moment. A branch manager who can see in real time that their branch is tracking well above last week's sales can motivate their team with that good news. A manager who can see that a particular menu item is underperforming can investigate and address the issue proactively rather than discovering it in a week-old paper report. For cashier staff, the transition from manual to digital transaction processing will meaningfully reduce the cognitive load of manual arithmetic under pressure — a genuine source of workplace stress that a well-designed digital POS system can eliminate.

For the Preservation of Ilonggo Food Heritage

Luna's Arrozcaldo is not just a business — it is a living piece of Ilonggo cultural heritage. Its recipes, its service philosophy, and its place in the community life of Iloilo City represent a form of cultural wealth that is, in the most literal sense, irreplaceable. By helping Luna's Arrozcaldo operate more efficiently and more sustainably, this research contributes, in a small but real way, to the preservation of that cultural wealth. A business that is well-managed, financially healthy, and able to compete effectively in the modern food service market is a business that can continue to serve its community for another 57 years and beyond.

Iloilo City's recognition as a UNESCO Creative City of Gastronomy in 2024 has brought renewed international attention to the city's extraordinary food culture. Luna's Arrozcaldo, as one of the city's most enduring and beloved food institutions, has a role to play in the story that Iloilo tells the world about its culinary heritage. A business that is thriving and well-organized is better positioned to welcome food tourists, participate in culinary events and festivals, and collaborate with other Ilonggo food institutions on initiatives that celebrate the region's culinary identity.

For the Broader Ilonggo MSME Community

The challenges that this study addresses are not unique to Luna's Arrozcaldo. They are the common challenges of the hundreds of family-owned food businesses that operate throughout Iloilo City and the wider Western Visayas region — businesses that have survived and thrived for decades on the strength of their food and their community relationships, but that now face increasing competitive pressure from larger, more digitally sophisticated operators. By demonstrating that a modest, well-designed digital system can meaningfully address these challenges, and by documenting the design and implementation process in accessible, replicable detail, this study makes a contribution to the digital transformation capacity of the broader MSME community. The Department of Trade and Industry's MSME Development Plan for 2023-2028 specifically identifies digital adoption as a key priority for Philippine MSMEs, and practical studies like this one are exactly the kind of knowledge resource that transformation requires.

For BSIT Students and the Academic Community

This study represents a sincere effort to demonstrate what capstone research in Information Technology can and should accomplish: a rigorous, technically sophisticated engagement with a real-world problem, in service of a real community, with lasting, practical outcomes. The researchers hope that this work will serve as a model and an inspiration for future BSIT students who are looking for ways to make their capstone research genuinely meaningful. The theoretical framework, system architecture, and evaluation methodology developed in this study are all designed to be rigorous enough to withstand academic scrutiny and accessible enough to be useful to practitioners. Future researchers who wish to extend this work — adding mobile functionality, integrating machine learning for sales forecasting, or expanding to additional MSME contexts — will find in this study a solid and well-documented foundation on which to build.

For Local Government and Tourism Stakeholders

Iloilo City's designation as a UNESCO Creative City of Gastronomy carries both a distinction and a responsibility. The city's tourism authorities and local government units have a shared interest in ensuring that heritage food institutions like Luna's Arrozcaldo remain operationally healthy and visible to visitors. A Luna's Arrozcaldo that is efficient, financially stable, and capable of handling high visitor volumes smoothly is a more effective ambassador for Iloilo's food culture than one struggling with paper receipts and manual inventory counts. Digital systems that improve service consistency and enable customer loyalty programs directly enhance the visitor experience in ways that benefit the broader tourism ecosystem.

For Technology Educators and Curriculum Developers

This study has implications for educators in information technology education in the Philippines. The challenge of designing digital systems for MSME clients — with limited budgets, mixed technical literacy, and deep cultural commitments to existing ways of working — is one many BSIT graduates will face. This study demonstrates a research and development approach that can serve as a model for future capstone projects, making the methodological choices — the use of Agile methodology, user acceptance testing protocols, and the framework for balancing technical ambition with client appropriateness — available for reflection and adaptation by future researchers.

Scope and Limitations of the Study

This research focuses on the design, development, and evaluation of a Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo, a multi-branch food enterprise with eight operational locations across Iloilo City. The study was conducted during the academic year 2025-2026, with preliminary data gathering initiated in January 2026 and system development and testing completed by June 2026. The primary client contact and main research informant is the current business owner of Luna's Arrozcaldo, who provided access to operational data, staff workflows, and business performance records essential to the research.

<u>Scope of the Study</u>

The proposed system encompasses the following functional domains and is designed for deployment across all eight Luna's Arrozcaldo branches: User Account Management and

Role-Based Access Control (supporting Cashier, Branch Manager, and Administrator roles); a Digital Point-of-Sale Terminal optimized for the high-speed food service counter environment; a Real-Time Inventory Management Module with automated threshold alerts and complete audit trails; a Centralized Sales Analytics and Reporting Dashboard presenting performance data in visual, filterable formats; a Customer Relationship Management feature supporting digital customer profiles and purchase history; a Promotions and Coupon Management Module with redemption tracking; and Staff Management Support through login activity logs and per-cashier transaction records.

The system is built on a technology stack comprising PHP for the backend application logic, PostgreSQL hosted on Supabase for the centralized database, HTML/CSS/JavaScript for the frontend user interface, Bootstrap for responsive UI components, Docker for containerized deployment, and Google OAuth for optional social authentication. The system supports eight branch locations: Festive Walk Mall, SM City Iloilo (Mandurriao), General Luna Street, Jaro, Molo, La Paz, Calumpang, and Tagbak. The system supports three distinct user roles, each with precisely defined access permissions: Administrator (full network-wide access), Branch Manager (branch-level access), and Cashier (POS terminal and basic inventory viewing only).

Limitations of the Study

The following boundaries have been established to ensure that the proposed system is achievable within the timeframe and resources of the capstone research program, and to focus development effort on the features that will deliver the greatest operational value to Luna's Arrozcaldo in the near term.

The proposed system does not include accounting or financial management functions beyond the recording and reporting of sales revenue. It does not perform double-entry bookkeeping, compute tax liabilities, generate financial statements, or interface with any external accounting software.

The system does not include payroll processing, time and attendance tracking, or formal human resources management. While the system records user login and logout activity and maintains cashier transaction records, it does not compute wages, generate payslips, or manage leave requests.

The system does not include a customer-facing online ordering interface, whether via web, mobile application, or self-service kiosk. All transactions in the proposed system are mediated by cashier staff using the POS terminal. Online ordering and delivery integration are identified as potential features for future development iterations.

The system is not optimized for smartphone screens and does not include a native mobile application for iOS or Android. The web-based interface is accessible from any device with a browser but is designed for desktop monitor sizes consistent with the counter-based workstations used at Luna's branches.

The system assumes reliable internet connectivity at each branch location. Integration with external payment gateways, delivery platforms, or third-party accounting systems is outside the scope of this study. The system handles cash and basic digital payment recording but does not process card payments directly or interface with e-wallet platforms.

Definition of Terms

The following terms are defined as they are used throughout this study. Where a term has both a general technical meaning and a specific application within the proposed system for Luna's Arrozcaldo, both meanings are provided. Terms are listed alphabetically for ease of reference.

Agile Development Methodology

A software development approach characterized by iterative development cycles (called sprints), close collaboration between developers and end users, and a flexible, adaptive response to changing requirements. In the context of this study, Agile methodology was chosen because it is particularly well suited to the development of systems with real, living clients whose operational needs may be better understood as development progresses and who benefit from seeing and providing feedback on working software early in the development process.

Analytical Reporting

The process of collecting, aggregating, and presenting business data in formats that support informed decision-making. Analytical reporting goes beyond simple data display to identify patterns, trends, and anomalies that would not be visible in raw data. In this study, analytical reporting refers specifically to the suite of reports, dashboards, and visualizations generated by the Digital Sales Processing System — including daily sales summaries, branch performance comparisons, menu item rankings, and hourly sales distribution maps.

Arrozcaldo

A Filipino rice porridge prepared with short-grain or glutinous rice, simmered in a rich chicken or beef broth seasoned with ginger, garlic, onion, and fish sauce, and garnished with fried garlic, green onions, and a boiled or poached egg. Luna's Arrozcaldo's version is distinguished by its use of native Ilonggo chicken, locally sourced ginger, and a broth recipe that has been prepared according to the same core formula since the business's founding in 1967. In the Ilonggo cultural context, arrozcaldo is strongly associated with comfort, care, and community.

Carinderia

A small, informal food establishment common throughout the Philippines, typically family-owned, offering inexpensive home-style cooked meals served counter-style or cafeteria-style. Carinderias form an essential part of the urban Philippine food ecosystem, serving as affordable dining options for workers, students, and lower-income households. Luna's Arrozcaldo began as a carinderia and retains the ethos of the carinderia tradition — unpretentious, generous, community-oriented — even as it has grown into a multi-branch enterprise.

Centralized Sales Management

A data management architecture in which sales records, inventory data, customer information, and operational logs from all branch locations are stored in and managed through a single, unified database system. Centralized management eliminates the fragmentation and inconsistency that result when each branch maintains its own separate records, and provides management with a single authoritative view of business performance across the entire operation. The system adopts a centralized architecture hosted on Supabase PostgreSQL as its foundational design principle.

Data Privacy Act of 2012 (Republic Act 10173)

The principal data privacy legislation of the Philippines, which establishes the legal framework for the collection, storage, use, and sharing of personal information in both the public and private sectors. The Act establishes the National Privacy Commission as the regulatory authority for data privacy in the Philippines and sets out the rights of data subjects, the obligations of personal information controllers and processors, and the penalties for violations. The proposed system is designed in full compliance with the requirements of the Data Privacy Act and its implementing rules and regulations.

Docker

An open-source platform for developing, shipping, and running applications in containerized environments. Containers are lightweight, standalone, executable packages that include everything needed to run a piece of software: code, runtime, system tools, libraries, and settings. In this study, Docker is used to package the system and its dependencies into a portable container that can be deployed consistently across different hosting environments, ensuring that the system runs identically in development, testing, and production.

<u>Ilonggo</u>

A term referring to the people, language, and culture of the Iloilo region of the Western Visayas, Philippines. As an adjective, "Ilonggo" describes something as originating from or characteristic of this region — hence "Ilonggo food," "Ilonggo cuisine," and in the context of this study, "Ilonggo Legacy," as used in Luna's Arrozcaldo's brand identity. The Ilonggo people are speakers of Hiligaynon and are known throughout the Philippines for their warm hospitality, refined culinary tradition, and distinctive cultural identity.

<u>Inventory Management Module</u>

The inventory management component responsible for tracking, controlling, and reporting on the quantities of products available at each Luna's Arrozcaldo branch. The module maintains real-time stock levels updated automatically as sales are processed through the POS terminal, supports manual adjustment entries for deliveries, spoilage, and inter-branch transfers, generates automated low-stock alerts when configurable thresholds are crossed, and maintains a complete audit trail of all inventory events. Each inventory event is logged with a timestamp and the user ID of the staff member who recorded it.

<u>MSME (Micro, Small, and Medium Enterprise)</u>

A classification of business enterprises defined in the Philippines by Republic Act 9501 (Magna Carta for MSMEs) based on asset size and number of employees. Micro enterprises have assets not exceeding ₱3 million; small enterprises have assets between ₱3 million and

₱15 million; medium enterprises have assets between ₱15 million and ₱100 million. Luna's Arrozcaldo is classified as a small to medium enterprise. MSMEs constitute approximately 99.5 percent of all business establishments in the Philippines and are a critical driver of employment and economic development, particularly in regional cities like Iloilo.

PHP (Hypertext Preprocessor)

A widely-used, open-source server-side scripting language designed primarily for web development. PHP code is executed on the server side, generating dynamic HTML content that is sent to the client's browser. In the proposed system, PHP handles all backend application logic: user authentication and session management, transaction processing, database queries, inventory calculations, report generation, and API responses. PHP was chosen for this project because it is widely understood by Filipino developers, well-documented, and compatible with the PostgreSQL database used by Supabase.

Point-of-Sale (POS) Terminal

A hardware and software system used to process sales transactions at the point where a customer purchases goods or services. In this study, the POS is a web-based software application running on a desktop or laptop computer at each Luna's Arrozcaldo branch counter. The POS interface presents a touch-friendly menu of all available products organized by category, enables cashiers to add items to orders, apply discounts and promotions, process payments, and generate digital receipts. Every transaction processed through the POS is automatically recorded in the central database and reflected immediately in the management dashboard.

PostgreSQL

An advanced, open-source object-relational database management system known for its reliability, feature robustness, and compliance with SQL standards. In the proposed system, PostgreSQL is used as the backend database, hosted on Supabase's cloud infrastructure in the

Asia-Pacific Southeast region for minimal latency from Iloilo City. The database stores all system data — users, branches, products, transactions, transaction items, customers, and password reset tokens — in a normalized relational structure optimized for the analytical queries required by the reporting module.

Role-Based Access Control (RBAC)

A security and access management framework that restricts each user's access to system features and data based on the role they have been assigned within the organization. RBAC is a widely-used security best practice in enterprise and business information systems. In the proposed system, RBAC is implemented through three defined roles — Cashier, Branch Manager, and Administrator — each with a specific, non-overlapping set of permissions determined by their operational responsibilities.

Sugilanon (Research Group Name)

The Hiligaynon word for "story" or "narrative." The name was chosen for this system to reflect the research team's conviction that business data is, at its core, a form of storytelling — that the transactions, inventory movements, and customer interactions recorded by the system together constitute the daily story of how Luna's Arrozcaldo serves its community. The name also honors the Ilonggo linguistic heritage of the business and its community, asserting that digital transformation need not mean the erasure of local cultural identity.

Supabase

An open-source Backend-as-a-Service (BaaS) platform built on PostgreSQL, providing a managed database, authentication, storage, and API generation service. Supabase is used in this study to host the central database, providing reliable cloud PostgreSQL hosting with built-in connection pooling via PgBouncer, SSL encryption, and a web-based SQL editor for database administration. The Supabase project is hosted in the AWS Asia-Pacific Southeast region, ensuring low-latency database connections from Iloilo City.

Web-Based System

A software application that runs on a web server and is accessed through a standard internet browser rather than being installed locally on each user's device. Web-based systems are particularly well suited for multi-branch businesses because they can be accessed from any internet-connected device, are centrally maintained and updated, and eliminate the need for complex local software installation and management. The proposed system is implemented as a web-based platform, making it accessible from any device with a modern browser at any Luna's Arrozcaldo branch or remote management location.

User Acceptance Testing (UAT)

A structured testing methodology in which representative end users evaluate a system's functionality, usability, and appropriateness against their actual operational needs before it is accepted for production deployment. UAT evaluates whether the system works as users actually need it to work — particularly important in contexts where users have limited technical backgrounds and translation between user requirements and technical specifications involves inherent interpretation gaps. In this study, UAT was conducted with representative cashier, branch manager, and administrator users at Luna's Arrozcaldo before the system was delivered to the client organization.

Bootstrap

An open-source front-end framework for developing responsive, mobile-first web interfaces. Bootstrap provides pre-built CSS styles and JavaScript components — navigation bars, buttons, forms, modals, and data tables — enabling developers to create consistent, professional-looking interfaces without writing all styling code from scratch. In this study, Bootstrap version 5 styles the front-end user interface of the Digital Sales Processing System, providing a clean, consistent visual design across all modules and ensuring the interface renders correctly on the desktop monitors used at Luna’s Arrozcaldo branch counters.

Theoretical Framework

The theoretical framework of this study is built on three intellectual pillars that together provide a rigorous and comprehensive foundation for the research: General Systems Theory, Digital Transformation Theory, and the Cultural Economics of Food Heritage Businesses. These three bodies of knowledge are not merely decorative academic scaffolding — they have actively shaped the research questions, design decisions, and interpretive approaches that characterize this study.

1. General Systems Theory and Business Information Systems

General Systems Theory (GST), developed by the Austrian biologist Ludwig von Bertalanffy and elaborated by subsequent scholars across many disciplines, provides the foundational lens through which this study views the Luna's Arrozcaldo business organization and the proposed information system. The core insight of GST is that complex phenomena — whether biological organisms, social organizations, or technological systems — are best understood not as collections of independent parts but as integrated wholes in which every element affects and is affected by every other element.

Applied to Luna's Arrozcaldo, this perspective reveals that the business's operational challenges are not isolated, independent problems but interconnected symptoms of a single systemic issue: the absence of a unified information infrastructure that connects all parts of the organization into a coherent, intelligible whole. The manual travel burden exists because there is no information infrastructure allowing the owner to see what is happening at each branch without physically visiting. The paper record problems exist because there is no digital channel through which transaction data can flow automatically from the point of sale to the management level. The oversight gaps exist because there is no system connecting branch-level events to centralized monitoring tools.

The proposed Digital Sales Processing System is, in GST terms, a new information infrastructure designed to connect all elements of the Luna's Arrozcaldo enterprise into a coherent, mutually reinforcing whole. When a cashier processes a transaction at a branch POS terminal, that single event simultaneously updates the branch sales record, reduces the branch inventory, adds to the customer's purchase history, and feeds the network-wide analytics dashboard. Every part of the system is connected to every other part, and information flows freely and instantly throughout the whole. This study also draws on the sub-discipline of Management Information Systems (MIS), which applies systems thinking specifically to the design of information systems in organizational contexts.

2. Digital Transformation Theory

The concept of digital transformation — the integration of digital technology into all areas of a business, fundamentally changing how it operates and delivers value — is central to understanding what this study attempts to accomplish. The theoretical literature on digital transformation, which has grown enormously over the past decade, offers a rich set of concepts and frameworks for thinking about how organizations transition from analog to digital modes of operation.

Westerman, Bonnet, and McAfee (2014), in their landmark study of digital transformation in established enterprises, draw a useful distinction between "digitization" — the use of digital tools to automate or accelerate existing processes without fundamentally changing them — and "transformation" — the use of digital tools to create genuinely new capabilities and business models. For Luna's Arrozcaldo, the proposed system represents primarily digitization in the first phase: it automates and improves existing processes (sales recording, inventory tracking, performance monitoring) rather than inventing entirely new business models. However, the analytical capabilities of the proposed system create the foundation for genuine transformation in subsequent phases, as data-driven insights enable management to make decisions about menu development, pricing, staffing, and marketing that would previously have been impossible without the underlying data.

The Technology Acceptance Model (TAM), originally developed by Davis (1989) and extensively refined since, provides a theoretical framework for predicting whether the proposed system will actually be adopted and used effectively by its intended users. TAM posits that technology adoption is primarily determined by two factors: perceived usefulness (the degree to which users believe the system will improve their work performance) and perceived ease of use (the degree to which users believe the system will be easy to use without excessive effort). The user interface design of the proposed system has been guided throughout by both of these TAM factors, with particular attention to ensuring that the POS interface — which will be used by cashier staff with varying levels of technical experience — is as simple and intuitive as possible.

3. Cultural Economics of Food Heritage Businesses

The third theoretical pillar of this study draws from the interdisciplinary field of cultural economics, which examines the economic dimensions of cultural goods, industries, and institutions. Food scholars including Katarzyna Cwiertka, Richard Wilk, and Sidney Mintz have argued persuasively that traditional food businesses are not merely commercial enterprises but repositories of cultural knowledge, social memory, and community identity — what economists call "cultural capital" in the sense developed by Pierre Bourdieu.

For Luna's Arrozcaldo, this cultural economic perspective has important practical implications. It suggests that the business's greatest competitive advantage is not its location, pricing, or even its operational efficiency — it is the cultural authenticity and community trust it has accumulated over 57 years. This cultural capital is both a valuable asset and a fragile one: it can be undermined by changes in food quality, by perceived abandonment of the business's community-oriented values, or by a technology implementation that feels cold, impersonal, or alien to the business's established character. This theoretical awareness has guided the researchers throughout the design of the system, particularly in decisions about user interface aesthetics and the framing of the system's purpose to the client and staff.

4. Human-Computer Interaction and Usability Theory

The fourth theoretical pillar draws from Human-Computer Interaction (HCI) and usability engineering. HCI is concerned with the design and evaluation of interactive computing systems for human use. The foundational work of Jakob Nielsen, Donald Norman, and Ben Shneiderman has established design principles widely adopted in professional software development. For the Luna’s Arrozcaldo system, HCI theory has particular importance because cashiers who will use the POS terminal daily are food service workers with limited experience with formal business software. Norman’s concept of “affordances” — the visual cues in an interface that suggest how it should be used — has been applied throughout the POS design to ensure the correct action in any situation is also the most visually obvious action.

The intersections between these four theoretical frameworks — General Systems Theory, Digital Transformation Theory, Cultural Economics, and Human-Computer Interaction — reflect a deliberate conviction that designing effective digital systems for MSME clients requires engagement with multiple bodies of knowledge simultaneously. A technically well-constructed system that ignores cultural context will fail to be adopted. A culturally sensitive system without sound information architecture will fail to deliver business value. Only by holding all four perspectives in productive tension can a development team produce a system that is technically sound, practically useful, culturally appropriate, and genuinely usable.

Conceptual Framework

The conceptual framework of this study translates the theoretical perspectives described in the preceding section into a practical operational model that has guided every phase of the proposed system's design and development. It is organized around the Input-Process-Output (IPO) model of information systems, enriched with the feedback loops that make the proposed system not merely a passive recorder of business events but an active contributor to organizational learning and continuous improvement.

Input: The Raw Material of Business Intelligence

The inputs to the proposed system are the raw operational data generated by Luna's Arrozcaldo's daily business activities across all eight branches. These inputs flow into the system through four primary channels: the POS terminal (which captures transaction data in real time as sales are processed), the inventory management interface (through which staff record deliveries, adjustments, and transfers), the customer management interface (through which staff create and update customer profiles), and the user management interface (through which administrators create and manage user accounts and permissions).

The quality of these inputs is fundamental to the value of everything the system produces. A system is only as good as the data it contains, and the reliability of analytical reports, inventory alerts, and customer insights depends entirely on the accuracy and completeness of the data entered at the source. For this reason, the system's input interfaces are designed with extensive data validation — preventing the entry of obviously incorrect values, requiring the completion of mandatory fields, and prompting users to confirm unusual entries before they are recorded. Training materials are designed to help staff understand not just how to use the system but why accurate data entry matters for the quality of the information the system provides to management.

Process: Transforming Data into Information

The processing layer of the system is where raw input data is transformed into meaningful, actionable information. This transformation happens at two levels: the database level, where data is stored, organized, and related according to the system's entity relationship model in PostgreSQL; and the application level, where business logic implemented in PHP is applied to produce the outputs — reports, alerts, dashboards — that management and staff interact with.

At the database level, the system's normalized, relational structure enables efficient retrieval and flexible querying. Sales transactions are linked to the items they contain, the customers who made them, the cashiers who processed them, and the branches where they occurred. Inventory records are linked to the products they track and to the events that have affected their levels. Customer records are linked to their transaction histories. This relational structure makes it possible to answer complex business questions — which branch has the highest average transaction value on weekday mornings? which menu items are most commonly purchased together? — with a single database query rather than hours of manual analysis.

Output: Information That Drives Action

The outputs of the system — the management dashboard, branch performance reports, inventory alerts, customer insights, and promotional analytics — are designed not merely to be accurate but to be actionable. The management dashboard presents the metrics that the business owner has identified, through preliminary interviews with the research team, as most important for daily operational decision-making: today's network-wide sales total relative to yesterday and last week's equivalent day; the branch with the highest and lowest current performance; any active inventory alerts; and a quick-view of any unusual activity flags. From this default view, the owner can drill down into any aspect of the data they wish to explore further, generating detailed reports on demand.

Feedback: The Organizational Learning Loop

The conceptual framework's most powerful feature is the feedback loop that connects the system's outputs back to its inputs and processes. As management uses the system's analytical outputs to make decisions — adjusting menu offerings, changing ingredient sourcing decisions, reallocating staff between branches, launching promotional campaigns — those decisions generate new operational events that flow back into the system as new inputs. Over time, this creates an organizational learning loop: the system's historical data becomes richer and more complete, the analytical insights it can generate become more nuanced and reliable, and the quality of the decisions made on the basis of those insights improves correspondingly. The proposed system is designed with this learning loop explicitly in mind, and will become more valuable to the business the longer it is used.

Security and Data Integrity Architecture

Underlying the Input-Process-Output architecture of the proposed system is a security and data integrity layer ensuring reliability, confidentiality, and auditability of all data. At the network level, all data is encrypted using Transport Layer Security (TLS). At the application level, passwords are stored as salted cryptographic hashes using PHP's bcrypt implementation. At the database level, PostgreSQL's transaction atomicity ensures each POS transaction is either fully recorded or fully rejected — there is no intermediate state in which a payment is processed but the corresponding inventory deduction fails. Every inventory event and login action is stored with a timestamp and user ID, providing a complete audit trail enabling management to trace any discrepancy to its source.

System Integration and Modular Architecture

The modular structure of the system — with each major functional domain implemented as a distinct module with clearly defined interfaces — means that individual modules can be enhanced or extended independently without disrupting others. A future version might replace the web-based POS terminal with a touch-screen tablet interface, or add a mobile application for the owner's management dashboard, without requiring changes to the inventory or reporting modules. The database schema's normalization and the use of UUID primary keys facilitate future integration with external platforms without schema refactoring, reflecting the system's design philosophy: not just to serve today's needs but to grow as the organization's capabilities expand.

Research Locale and Client Profile

This study was conducted at the multiple branch locations of Luna's Arrozcaldo across Iloilo City, with the primary research locale being the La Paz branch — the original location established by the founder in 1967 — and the management office used by the current business owner for record-keeping and administrative activities. Iloilo City is the capital of the Province of Iloilo and the regional center of Western Visayas (Region VI) in the Philippines. It is the sixth most populous city in the Philippines, with a population of approximately 477,000 (2020 census), and is recognized as one of the country's most livable and culturally rich urban centers.

Iloilo City's food culture is one of its most distinctive and celebrated attributes. The city has been designated a UNESCO Creative City of Gastronomy since 2024, recognizing its extraordinary culinary heritage, the depth of its food traditions, and the vibrancy of its contemporary food scene. The designation acknowledges Iloilo's unique positioning at the intersection of indigenous Visayan culinary traditions, Spanish colonial culinary influences, and modern Filipino food innovation. Establishments like Luna's Arrozcaldo are central to the living food culture that earned Iloilo this international recognition.

Luna's Arrozcaldo currently operates eight branches within Iloilo City, each serving the residential and commercial community of its surrounding district. The Festive Walk Mall branch serves the shopping and entertainment complex in Mandurriao district. The SM City Iloilo branch, also in Mandurriao, serves one of the city's largest shopping destinations. The General Luna Street branch serves the commercial and heritage district of downtown Iloilo. The Jaro branch serves the historic Jaro district, known for its basilica and longstanding residential community. The Molo branch serves the Molo district, another historic area with a strong community identity. The La Paz branch is the original location and serves the surrounding working-class neighborhood that has supported Luna's since 1967. The Calumpang branch serves a growing residential and commercial area. The Tagbak branch, located in the Tagbak area of Jaro, serves a transit-adjacent community.

Each branch operates with a team of cashiers under the supervision of a branch manager who reports directly to the business owner. Product offerings are standardized across all branches, with the core arrozcaldo menu supplemented by additional rice meals, breakfast items, merienda offerings, beverages, and desserts. Branches serve customers from early morning — opening hours typically begin at 6:00 AM to serve the breakfast crowd — through the late afternoon or evening depending on location and customer demand patterns.

The current generation of management, consisting of family members who grew up in the business and have taken on increasing responsibility as they reached adulthood, is characterized by a blend of deep respect for the business's traditions and a practical awareness of the need to adapt to contemporary conditions. The primary owner-manager, who serves as the main client contact for this research, demonstrated throughout the research engagement a sophisticated understanding of both the business's strengths and its operational challenges, and expressed genuine enthusiasm for the transformative potential of the proposed digital system while maintaining clear-eyed expectations about what a first-phase digital system can and cannot accomplish.

The research team conducted structured interviews with the owner-manager, two branch managers (from the La Paz and Festive Walk branches), and three cashier staff members over the preliminary research period. These interviews provided the detailed operational data — transaction volumes, shift workflows, inventory management practices, staff training approaches — that informed both the problem statement and the system design described in the subsequent chapters of this report. The team also conducted on-site observation sessions at two branches, watching the end-of-shift reconciliation process and the morning setup routine that confirmed many of the inefficiencies documented in the statement of the problem.

The research team is grateful to the Luna family and the branch staff for their openness, patience, and genuine partnership throughout this project. The quality of the information they shared, and the trust they extended to four BSIT students from PHINMA University of Iloilo, made this research richer, more grounded, and more practically valuable than it could otherwise have been.

Branch-Level Operational Observations

The research team's on-site observation sessions at the La Paz and Festive Walk Mall branches provided a granular understanding of operational realities inaccessible through interviews alone. At the La Paz branch, the research team observed the full end-of-shift reconciliation process: the cashier totaling the day's order slips by hand, transferring subtotals to a summary sheet, then reconciling the cash drawer against the paper total. The process took approximately 45 minutes, followed by 15 to 20 minutes of independent review by the branch manager — over an hour of productive time consumed at the end of every shift by a process the digital POS system would reduce to minutes.

Client Engagement and Co-Design Process

The research team's engagement with Luna's Arrozcaldo went beyond data collection to encompass a genuine co-design process guided by Agile methodology. After each sprint, the team presented working prototypes to the owner-manager and branch managers, generating feedback that directly influenced subsequent development. The initial POS interface organized menu items in a vertical list — an approach the branch manager feedback revealed was poorly suited to the counter environment. The interface was redesigned after Sprint 1 to prominently feature the most popular items. Similarly, the initial inventory alert system's color-only metaphor was replaced after Sprint 2 feedback with explicit text labels displaying current stock level, threshold level, and a direct link to the order entry screen.

Publication and Ethics Declaration

This capstone project report is published as part of the PHINMA University of Iloilo College of Information Technology Education Capstone Research Series, a peer-reviewed institutional publication assigned Print ISSN 2799-4821, Online ISSN 2799-4839, and ISBN 978-621-96482-3-7 by the Research and Publications Office of PHINMA University of Iloilo.

The researchers declare that this study was conducted in accordance with the ethical standards of academic research and the specific research ethics guidelines of PHINMA University of Iloilo. Informed consent was obtained from all research participants prior to the conduct of interviews and observation sessions. Participants were informed of the nature and purpose of the research, their right to decline participation or withdraw at any time without consequence, and the measures taken to protect the confidentiality of any sensitive business information they shared with the research team.

The business data shared by Luna's Arrozcaldo in the course of this research — including sales figures, inventory records, and operational workflow descriptions — is presented in this report only in aggregated, anonymized, or illustrative form where necessary to protect the business's competitive interests. Specific financial figures, supplier names, and pricing information have been generalized or omitted in accordance with a confidentiality agreement negotiated with the client organization prior to the commencement of data collection.

All customer data collected or referenced in the development and testing of the system's customer management module was handled in compliance with the Data Privacy Act of 2012 (Republic Act 10173) and its implementing rules and regulations. No real customer personal data was used in system testing; all test data was synthetically generated. The production deployment of the system includes appropriate privacy notices, data minimization practices, and access controls as required by the Act.

The source code of the system, the database schema, and the technical documentation developed in this study are the joint intellectual property of the research team and PHINMA University of Iloilo, licensed to Luna's Arrozcaldo for operational use under the terms of the

Capstone Research Intellectual Property Agreement executed between the parties at the commencement of this research. Third-party libraries and frameworks used in the development of the system — including Bootstrap, Chart.js, Font Awesome, and the Supabase client libraries — are used in accordance with their respective open-source licenses.

The researchers affirm that this study is original work and has not been submitted for academic credit at any other institution. All external sources cited in this report are properly attributed in accordance with the citation standards of the College of Information Technology Education at PHINMA University of Iloilo. The researchers take full responsibility for the accuracy and integrity of the information presented in this report.

— End of Chapter I —

Design and Development of a Digital Sales Processing System
for Luna's Arrozcaldo — Ilonggo Legacy Since 1967
PHINMA University of Iloilo · Data Informatics Track · A.Y. 2025–2026

PHINMA UNIVERSITY OF ILOILO

COLLEGE OF INFORMATION TECHNOLOGY EDUCATION

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FOR LUNA'S ARROZCALDO — ILONGGO LEGACY SINCE 1967

A Capstone Project

Presented to the Faculty of the
College of Information Technology Education
PHINMA University of Iloilo

In Partial Fulfillment
Of the Requirements for the degree
Bachelor of Science in Information Technology

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TABLE OF CONTENTS

CHAPTER II

REVIEW OF RELATED LITERATURE

This chapter presents the related literature and studies relevant to the Design and Development of a Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo. The literature reviewed covers seven thematic areas closely aligned with the core components of the proposed system: Point-of-Sale and digital transaction processing systems, digital transformation in micro, small, and medium enterprises (MSMEs), centralized data management and business intelligence in food service operations, inventory management systems for multi-branch food businesses, customer relationship management and loyalty programs, role-based access control and information security, and the cultural economics of heritage food institutions in the Philippine context. Together, these bodies of literature provide the theoretical and empirical foundation upon which the proposed system was designed and the research conducted.

The researchers acknowledge that the literature available on digital systems specifically for heritage food businesses in the Philippine regional context is limited, and that much of the directly applicable body of knowledge must be assembled from adjacent fields: food service management, MSME digitization, and information systems development. Where direct

Philippine examples are available, they are prioritized. Where relevant research originates from other national or international contexts, the researchers have taken care to identify the dimensions of applicability and the qualifications required when translating findings to the specific context of Luna's Arrozcaldo and Iloilo City.

The synthesis of these thematic areas converges on a central proposition that has guided every design decision in the proposed system: that the careful, culturally sensitive, and user-centered development of a digital information system for a heritage food MSME can produce not only technical functionality but genuine organizational value — improving efficiency, reducing errors, enhancing management visibility, and strengthening community relationships — while fully preserving the cultural authenticity and community spirit that define the institution's identity and sustain its competitive advantage.

POINT-OF-SALE SYSTEMS AND DIGITAL TRANSACTION PROCESSING

The Point-of-Sale (POS) terminal represents one of the most consequential digital touchpoints in any retail or food service business. It is the interface through which every revenue-generating transaction of the enterprise is initiated, recorded, and stored — the first link in the information chain that connects a customer order to the business's financial records, inventory levels, and management reports. The evolution of POS systems over the past three decades has tracked closely with the broader trajectory of enterprise computing: from standalone hardware-based cash registers, to client-server networked systems, to the cloud-based, browser-accessible platforms that represent the current state of the art.

Parsa and Khan (2011) documented the early adoption patterns of POS technology in the food service industry, noting that the primary drivers of POS adoption among restaurant operators were reductions in transaction processing time, improvements in order accuracy, and the elimination of arithmetic errors in shift-end reconciliations. These findings from over a decade ago are directly relevant to the problems documented at Luna's Arrozcaldo today: the business's current manual transaction recording process exhibits precisely the inefficiencies — slow processing, frequent arithmetic errors, laborious reconciliation — that POS systems are designed to address. The research team's direct observation of the end-of-shift reconciliation process at the La Paz and Festive Walk branches confirmed that the documented inefficiencies are not isolated incidents but systematic features of the current manual system.

The shift from hardware-centric POS terminals to web-based POS platforms has democratized access to sophisticated transaction processing technology for MSMEs that could not previously afford dedicated POS hardware. Wailgum (2013) noted that cloud-based POS systems eliminate the need for expensive server hardware and reduce the technical maintenance burden on small business operators, while providing functionality previously available only to large enterprises with dedicated IT departments. The proposed system for Luna's Arrozcaldo is designed in precisely this spirit: a web-based POS accessible from any modern browser-capable device, requiring no dedicated POS hardware beyond the counter workstations already present at each branch.

The design of POS user interfaces for food service environments presents specific challenges not present in general retail contexts. Counter staff in busy food service operations must process transactions rapidly under conditions of high sensory load — simultaneous customer interactions, kitchen noise, and the time pressure of queuing customers. Olsen and Connolly (2000) emphasized that POS interfaces in restaurant environments must minimize the number of taps or clicks required to complete a transaction, present the most frequently ordered items in the most visually prominent positions, and provide clear error feedback that allows cashiers to correct mistakes without re-entering entire orders. The research team's workflow observation sessions at Luna's Arrozcaldo's branches confirmed that the pace and sensory environment of the counter service context are precisely as demanding as described in the literature, reinforcing the importance of these interface design principles.

In the Philippine food service context, Barredo et al. (2023) documented the implementation of a digital ordering system for Bugana, an electronic marketplace for food producers in Bago City, Negros Occidental. Their study found that digitizing the transaction interface reduced order processing time by an average of 40 percent and reduced end-of-shift reconciliation discrepancies to near zero. While Bugana is primarily a consumer-facing marketplace rather than a branch-level POS system, the operational dynamics — food service workers with limited technical backgrounds using digital tools in high-speed service environments — closely parallel those at Luna's Arrozcaldo.

The integration of POS systems with backend inventory and reporting databases represents a significant architectural challenge in multi-branch deployments. Muller (2019) observed that POS systems that write transaction data to local databases and synchronize periodically with a central server are vulnerable to data consistency problems when synchronization fails — a common problem in environments with unreliable internet connectivity. The proposed system's architecture addresses this by using synchronous database writes to the Supabase-hosted PostgreSQL backend at the time of each transaction, accepting a dependency on internet connectivity in exchange for guaranteed data consistency across the entire branch network.

Kassim and Baharudin (2017) investigated POS implementation outcomes in small restaurant chains across Southeast Asia, finding that systems implemented with extensive staff involvement in the design phase — where cashier staff were invited to evaluate prototype interfaces and provide specific feedback — achieved significantly higher adoption rates and lower error rates in production use than systems imposed without staff consultation. This finding directly supports the Agile co-design methodology employed in this study, and the research team's experience confirms its validity: the most significant improvements to the proposed system's POS interface emerged from Sprint 1 and Sprint 2 feedback sessions with Luna's Arrozcaldo cashier staff rather than from the initial design specification.

POS SYSTEMS IN PHILIPPINE MSME FOOD BUSINESSES

The adoption of POS technology among Philippine MSMEs in the food service sector has been studied by several researchers in recent years, with findings that illuminate both the barriers to adoption and the transformative impact of successful implementations. Quimba and Calizo (2019) found that among Philippine food service MSMEs surveyed, the most commonly cited barriers to POS adoption were upfront cost, perceived complexity, and uncertainty about the return on investment. The proposed system for Luna's Arrozcaldo addresses these barriers through a web-based architecture that eliminates hardware costs and a user interface design validated through iterative user testing to ensure accessibility for staff without prior digital experience.

Gañgan (2025) documented the implementation of a web-based market goods trading system for the Agri-Pinoy Trading Center, finding that the shift from manual to digital transaction recording reduced end-of-day reconciliation time from an average of 52 minutes to 8 minutes — a reduction of 85 percent. Comparable results are anticipated for Luna's Arrozcaldo, where the current end-of-shift reconciliation process was observed by the research team to consume approximately 45 minutes per shift. Annualized across eight branches operating two shifts per day, the current reconciliation process consumes approximately 4,380 hours of staff time per year — time that could be substantially recovered through the proposed digital system.

The Philippine retail and food service landscape has experienced significant POS technology adoption acceleration in the five years since 2020, driven in large part by pandemic-era operational requirements and the expansion of GCash, Maya, and other digital payment platforms. The Department of Trade and Industry's 2022 survey of MSMEs found that POS adoption rates in the food service sector increased from 12 percent in 2019 to 31 percent in 2022 — a nearly threefold increase driven primarily by the need to accept cashless payments during and after the pandemic. However, the same survey found that the majority of adopting MSMEs were using simple, single-function POS systems without inventory or reporting integration — precisely the more comprehensive, integrated functionality that the proposed system delivers.

DIGITAL TRANSFORMATION IN MSME FOOD SERVICE BUSINESSES

The digital transformation of micro, small, and medium enterprises in the food service sector has emerged as a prominent area of academic and policy interest in the Philippines and throughout Southeast Asia, accelerated by the disruptions of the COVID-19 pandemic and the rapid expansion of digital infrastructure in the region. Digital transformation in the MSME context is generally understood as the integration of digital technology into business operations in ways that create new value and capabilities, rather than merely automating existing processes without fundamentally changing them (Westerman, Bonnet & McAfee, 2014).

The distinction drawn by Westerman et al. (2014) between digitization — the direct translation of existing analog processes into digital form — and digital transformation — the creation of genuinely new operational capabilities through digital technology — is important for understanding the scope and ambition of this study. In its initial phase, the proposed system primarily digitizes existing Luna's Arrozcaldo processes: sales recording is digitized through the POS module, inventory tracking is digitized through the inventory management module, and performance monitoring is digitized through the management dashboard. However, the analytical reporting capabilities of the system create the foundation for genuine transformation in subsequent phases, as data-driven insights enable management decisions about menu optimization, pricing, staffing, and marketing that would previously have been impossible without the underlying historical data.

Briones, Galang, and Latigar (2023), in their comprehensive assessment of digital technology adoption in Philippine agri-food systems, identified five key dimensions of digital transformation readiness among Philippine food MSMEs: leadership commitment, staff digital literacy, technology infrastructure, data management capacity, and access to implementation support. Their analysis found that most Philippine food MSMEs score well on leadership commitment — owners are generally aware of the importance of digital tools and express readiness to adopt them — but face significant challenges on the remaining four dimensions. This profile closely matches that of Luna's Arrozcaldo, whose owner demonstrated throughout the research engagement both a strong commitment to digital transformation and practical concerns about staff readiness and implementation complexity.

The Technology Acceptance Model (TAM), originally proposed by Davis (1989) and extensively validated across hundreds of subsequent studies, provides a well-established framework for predicting whether end users will accept and effectively use a new information system. TAM identifies two primary determinants of technology acceptance: perceived usefulness — the degree to which users believe the system will help them perform their work better — and perceived ease of use — the degree to which users believe they will be able to use the system without excessive difficulty. Research applying TAM to food service POS systems (Shin & Leem, 2011) has consistently found that ease-of-use perceptions are particularly important in environments where users have limited prior digital experience, as they strongly moderate the relationship between perceived usefulness and actual system use.

These TAM insights have directly shaped the user interface design philosophy of the proposed system. Every design decision — the organization of menu items by category and popularity frequency in the POS interface, the use of color-coded inventory alerts supplemented by explicit text labels, the default date range settings in the reporting module, the placement and labeling of action buttons — reflects a deliberate effort to maximize perceived ease of use. The iterative design process conducted with Luna's Arrozcaldo staff through the Agile development sprints allowed the research team to test TAM-informed design choices against real user behavior and refine the interface accordingly with each sprint cycle.

The impact of the COVID-19 pandemic on digital transformation among Philippine food MSMEs has been documented by multiple researchers. De Jesus (2017), in a study of e-commerce adoption among agribusiness MSMEs in Cavite, found that adoption was strongly correlated with the owner-manager's personal digital competence and their prior exposure to digital tools in non-business contexts. The pandemic's forced digital adoption — as businesses scrambled to establish online ordering capabilities or remote management systems during lockdowns — appears to have accelerated this personal competence development among Philippine food MSME operators in ways that make the current environment significantly more favorable for digital transformation initiatives than it was before 2020.

Andal-Ancion, Cartwright, and Yip (2020), writing specifically about the implications of COVID-19 for MSME digital transformation in Southeast Asia, argued that the pandemic had compressed what would otherwise have been a decade-long digital transition into a period of two to three years, creating both an urgent need and a heightened organizational readiness for digital tools. Their recommendations for supporting MSME digital transformation — prioritizing ease of use over feature richness, providing extensive training and implementation support, and framing technology adoption in terms of specific operational problems it solves — have been applied throughout the design and implementation of the proposed Luna's Arrozcaldo system.

AGILE METHODOLOGY IN MSME SYSTEM DEVELOPMENT

The selection of Agile development methodology for this study reflects a deliberate alignment between methodology and client context. Agile's core principles — iterative development, frequent client feedback, adaptive response to changing requirements, and the prioritization of working software over comprehensive documentation — are particularly well-suited to the development of digital systems for MSME clients who may have difficulty articulating their requirements in the abstract but can respond concretely and usefully to working prototypes.

Highsmith and Cockburn (2001), in their foundational articulation of Agile principles, identified client collaboration as the methodology's most critical success factor, arguing that the most valuable input to a software development project is not a detailed requirements document but ongoing feedback from users interacting with working software. This principle proved particularly applicable in the Luna's Arrozcaldo context, where the initial user interviews revealed a partial and sometimes inconsistent picture of operational requirements that was substantially clarified and corrected through feedback on working Sprint 1 and Sprint 2 prototypes. For example, the initial interview data suggested that inventory management was a secondary concern for cashier staff; Sprint 1 prototype testing revealed that cashiers were actually highly interested in real-time stock level visibility at the POS terminal itself, leading to the addition of a current-stock indicator to the POS interface in Sprint 2.

Sharma, Sarkar, and Gupta (2012) conducted a comparative study of Agile and Waterfall methodology outcomes in small-to-medium software development projects, finding that Agile projects consistently produced higher client satisfaction scores and lower rates of feature misalignment, particularly in projects where client requirements were complex, partially specified, or likely to evolve during development. Luna's Arrozcaldo's operational requirements exhibited all three of these characteristics: they were complex (spanning POS, inventory, customer management, and analytics), partially specified (the owner knew what problems to solve but not which technical features would best solve them), and likely to evolve (as the development team's growing familiarity with branch operations revealed operational nuances not visible in initial interviews). These conditions collectively make Agile the strongly preferred methodology for this study.

The Sprint cycle structure adopted in this study — two-week development sprints followed by client review sessions with the Luna's Arrozcaldo owner and branch managers — provided a cadence of regular feedback that kept development focused on genuine operational needs and prevented the accumulation of undetected misalignment between what the developers were building and what the client actually needed. Post-project reviews of Agile implementations in comparable contexts (Dingsøyr et al., 2012) have consistently identified the frequency and quality of client feedback as the strongest predictor of project outcome quality, reinforcing the value of the Sprint review process embedded in this study's methodology.

CENTRALIZED DATA MANAGEMENT AND BUSINESS INTELLIGENCE IN FOOD SERVICE

The shift from decentralized, branch-level data management to centralized database architectures represents one of the most significant operational transformations available to multi-branch food service businesses. In a decentralized system, each branch maintains its own records — whether on paper or in a local digital system — and summary information is periodically aggregated at the management level through manual collection and consolidation. In a centralized system, all data from all branches flows in real time to a single, unified database, from which management can generate comprehensive, accurate, and current views of the entire operation at any moment.

The operational and managerial advantages of centralized data architectures in multi-branch food service businesses have been documented across a range of research contexts. Kimes (2011), in a study of information technology adoption in chain restaurant operations, found that multi-branch operators who implemented centralized POS and management systems reported significant improvements in their ability to identify underperforming branches, detect operational anomalies in real time, and make evidence-based staffing and supply chain decisions. These findings are directly relevant to the management challenges at Luna's Arrozcaldo, where the absence of centralized data has created the informational blindness that currently requires the owner to make daily physical visits to each of the eight branches.

Negash (2004) provided an influential definition of business intelligence as systems that combine data gathering, data storage, and knowledge management with analytical tools to present complex internal and competitive information to planners and decision makers. The analytical reporting module of the proposed system is explicitly designed as a business intelligence tool: it does not simply display raw transaction data but transforms it into comparative performance metrics, trend analyses, and ranked item performance reports that allow management to identify patterns and make evidence-based decisions. The management dashboard's visual representation of branch performance comparisons, daily sales trend lines, and menu item popularity rankings are all business intelligence outputs in Negash's sense — synthesized, actionable information derived from the underlying transaction data.

The PostgreSQL database management system, which provides the backend for the proposed system's Supabase-hosted database, has been widely evaluated in the literature as a robust and scalable platform for operational business databases. Momjian (2001) documented PostgreSQL's ACID (Atomicity, Consistency, Isolation, Durability) transaction model, which ensures that each sales transaction processed through the POS is either fully committed to the database or fully rolled back in the event of a failure. This transaction reliability guarantee is particularly important in the multi-branch food service context, where simultaneous transactions from eight branches are processed against a shared database and any partial transaction recording would create inventory and financial reporting discrepancies.

The selection of Supabase as the cloud hosting platform for the system's PostgreSQL database reflects both technical and practical considerations. Supabase provides managed PostgreSQL hosting in the AWS Asia-Pacific Southeast region, ensuring minimal network latency from Iloilo City. Its built-in connection pooling through PgBouncer supports the concurrent database connections required when multiple branch POS terminals process transactions simultaneously. Its web-based administration interface reduces the technical barrier for ongoing database management. And its open-source licensing model eliminates the per-user or per-record licensing fees that characterize many commercial database platforms, making the system economically sustainable for an MSME client with a limited technology budget.

The normalization of the proposed system's database schema — organizing data into separate, related tables for users, branches, products, transactions, transaction items, customers, and inventory records — reflects established best practices for relational database design in operational business information systems. Date (2003) articulated the theoretical foundations of database normalization, demonstrating that properly normalized relational schemas eliminate data redundancy, prevent update anomalies, and ensure data integrity in ways that are particularly important when multiple users are simultaneously reading and writing to the same database. The entity relationship model developed for the proposed system has been designed to Third Normal Form (3NF), ensuring that these data integrity guarantees hold across all operational scenarios.

REAL-TIME REPORTING AND DASHBOARD DESIGN

The design of management dashboards for multi-branch food service operations has been addressed in a growing body of human-computer interaction and management information systems research. Few and Edge (2006) articulated influential principles for dashboard design, emphasizing that effective dashboards should present only the metrics that are genuinely decision-relevant, should use visual encodings — charts, color indicators, and comparative bars — that allow the viewer to assess performance status at a glance without detailed numerical analysis, and should provide clear pathways from summary-level dashboard views to the detailed data underlying each summary metric.

These principles have been applied throughout the design of the proposed system's management dashboard. The dashboard's default view is organized around the specific management questions identified through preliminary interviews with the Luna's Arrozcaldo owner as most important for daily operational oversight: how is today's network-wide sales total comparing to yesterday and to the same day last week? Which branches are tracking ahead of or behind their recent performance baselines? Are there any inventory alerts requiring immediate attention? Are there any active promotions? The answers to these questions are presented in visual, at-a-glance formats that allow the owner to assess the state of the entire business in under a minute — a dramatic improvement over the current process of physically visiting each branch over two to four hours to gather the same information.

Chaudhuri, Dayal, and Narasayya (2011) documented the evolution of business intelligence reporting tools from static scheduled reports to interactive, ad-hoc query-capable dashboards, noting that the shift to interactive reporting significantly expands the range of business questions that non-specialist users can answer independently without requesting custom reports from IT staff. The proposed system's analytical reporting module incorporates this interactive capability: management can filter sales reports by branch, date range, product category, and user role, generating ad-hoc analyses tailored to specific management questions without requiring any technical database knowledge.

Wixom and Watson (2010) identified several critical success factors for business intelligence implementations in organizational settings: executive sponsorship, a clear data quality governance strategy, user involvement in report design, and an incremental rollout approach that builds user confidence progressively. All four of these factors have been addressed in the proposed system implementation: the Luna's Arrozcaldo owner has been an active and enthusiastic champion throughout the development process; data quality is protected through input validation at the POS interface; branch managers participated in the design of branch-level reports; and the system rollout plan calls for initial deployment at one branch for parallel operation testing before full network deployment.

INVENTORY MANAGEMENT SYSTEMS FOR MULTI-BRANCH FOOD BUSINESSES

Inventory management is a critical operational function in any food service business, with direct implications for food cost, service quality, and waste reduction. The unique challenges of food inventory management — short shelf life, demand variability, the need for correct stock rotation, and the high cost of both stockouts and over-ordering — have generated a substantial body of research on inventory management systems specifically designed for the food service context.

Bhatt and Bhatt (2012) conducted a comprehensive review of inventory management practices in multi-branch food service operations, finding that businesses using manual inventory management experienced average food waste rates of 8 to 12 percent of total food cost, while businesses using digital inventory management with automated reorder alerts experienced waste rates of 3 to 5 percent — a reduction of more than 50 percent. These findings align closely with the inventory management challenges documented at Luna's Arrozcaldo, where branch managers described regular instances of both stockouts and over-ordering, both of which represent measurable financial costs that the proposed system's inventory module is specifically designed to address.

The implementation of threshold-based inventory alerts — automated notifications generated when stock levels fall below configurable minimum levels — represents one of the most impactful features a digital inventory system can provide to food service operators. Davis and

Hamilton (2016) documented the implementation of threshold alerts in a chain of quick-service restaurants, finding that the alerts reduced stockout incidents by 78 percent compared to the pre-digital manual monitoring system. The proposed system implements threshold-based alerts for all inventory items at each Luna's Arrozcaldo branch, with configurable thresholds allowing branch managers to set appropriate minimum levels for each item based on their specific demand patterns.

The relationship between POS transaction data and inventory management is a key architectural feature of the proposed system. Each sale processed through the POS terminal automatically deducts the corresponding ingredient or product quantities from the branch inventory — a process called real-time inventory deduction that eliminates the lag between a sale and its inventory impact that characterizes manual inventory management. Muller (2019) identified real-time inventory deduction as the single most valuable feature of integrated POS-inventory systems for food service businesses, because it creates a continuous, accurate picture of current stock levels that allows managers to anticipate shortfalls before they occur rather than discovering them when a stockout disrupts service.

The challenge of managing inventory across multiple branches introduces coordination complexities beyond those present in single-location operations. In multi-branch food service networks, inventory imbalances between branches — one branch with excess stock while another faces a shortfall — create opportunities for inter-branch transfers that can reduce waste and prevent service disruptions. The proposed system's inventory module includes inter-branch transfer recording as a core feature, creating a complete audit trail of all inventory movements and making it possible for the administrator to identify and coordinate inter-branch inventory rebalancing before imbalances become service problems.

The economic literature on optimal inventory management in food service businesses has extensively documented the cost asymmetry between stockout costs and overstock costs.

Chopra and Meindl (2016) showed that in food service businesses where items cannot be substituted and customer expectations of product availability are high, the cost of a stockout — in lost revenue, customer disappointment, and long-term loyalty erosion — consistently exceeds the cost of marginal overstock, suggesting that safety stock levels should be set conservatively. The proposed system's configurable threshold system allows branch managers to set their reorder alerts at levels that reflect this cost asymmetry, with the general recommendation that alert thresholds be set at twice the average daily consumption of each item.

TECHNOLOGY FOR FOOD WASTE REDUCTION

The reduction of food waste is a concern that transcends operational efficiency to encompass environmental sustainability and social responsibility — values deeply embedded in the community-oriented ethos of Luna's Arrozcaldo. The Philippines generates approximately 9 million metric tons of food waste per year according to Food and Agriculture Organization estimates, with the food service sector accounting for a disproportionate share. Research by the Philippine Council for Agriculture, Aquatic and Natural Resources Research and Development has identified improved inventory management and demand forecasting as the most impactful technological interventions available to food service businesses seeking to reduce waste.

Digital inventory systems that integrate real-time stock tracking with sales trend data can help food service operators make more precise purchasing decisions, reducing the over-ordering that leads to perishable ingredient spoilage. While the proposed system's initial version does not include an automated demand forecasting module, the historical transaction data it will generate over its first year of operation creates the foundation on which such a module can be built in a future development iteration — transforming the system from a reactive tool that records inventory events to a proactive tool that anticipates future demand based on identified sales patterns.

Parfitt, Barthel, and Macnaughton (2010), in a landmark analysis of global food waste patterns, identified inventory over-ordering at the retail and food service level as a more significant driver of food waste than household food disposal — a finding with clear implications for the operational priority that food service businesses should assign to inventory management improvement. For Luna's Arrozcaldo, where perishable ingredients including fresh chicken, ginger, and herbs constitute a significant proportion of total ingredient costs, the reduction in over-ordering waste enabled by the proposed system's real-time inventory tracking directly translates to cost savings and reduced environmental impact.

CUSTOMER RELATIONSHIP MANAGEMENT AND LOYALTY PROGRAMS IN FOOD SERVICE

Customer relationship management (CRM) — the systematic collection, analysis, and use of customer data to improve retention, increase visit frequency, and deepen customer loyalty — has become an increasingly important dimension of food service marketing strategy in the digital era. The proliferation of digital transaction channels has created unprecedented opportunities for food service businesses to capture and analyze customer behavior data that was previously invisible or inaccessible, enabling personalized marketing and retention strategies that were previously available only to large chain operators with dedicated marketing departments.

Peppers and Rogers (1993), in their foundational articulation of one-to-one marketing principles, argued that the competitive advantage of the future would belong to businesses that could treat different customers differently — identifying their most valuable customers and investing disproportionately in retaining and deepening those relationships. Applied to the food service context, this principle suggests that Luna's Arrozcaldo's most important competitive asset may not be its location or pricing but its ability to identify and reward the loyal customers who visit most frequently — customers who are, by definition, already deeply attached to the brand and highly responsive to recognition and reward.

Kivela, Inbakaran, and Reece (2000) conducted foundational research on customer loyalty in restaurant contexts, finding that the factors most strongly predictive of repeat visitation were food quality consistency, service warmth, and perceived value — all of which Luna's Arrozcaldo has cultivated over 57 years. However, their research also found that loyalty programs — formal mechanisms for recognizing and rewarding loyal customers — significantly amplified the relationship between these core satisfaction factors and actual repeat visitation rates. The proposed system's customer management module creates the digital infrastructure for loyalty program implementation: a database of customer profiles linked to purchase histories from which the most loyal customers can be identified and targeted with recognition and reward.

Alampay et al. (2019) documented the impact of a digital loyalty program implementation at a chain of casual dining restaurants in Metro Manila, finding that customers enrolled in the loyalty program visited on average 2.3 times per month compared to 1.1 times per month for non-enrolled customers — a visit frequency more than double that of the non-enrolled cohort. The researchers cautioned, however, that the most significant predictor of enrollment and engagement with the loyalty program was the ease of the enrollment process: customers who found enrollment cumbersome or confusing opted out at high rates. This finding has influenced the design of the proposed system's customer profile creation feature, which has been streamlined to require only the minimum information necessary for loyalty program administration.

The Data Privacy Act of 2012 (Republic Act 10173) establishes important constraints on the collection and use of customer personal data by Philippine businesses, including food service operators implementing loyalty programs. The Act requires that personal data be collected only for specified, legitimate purposes; that customers be informed of data collection and provide consent; that data be stored securely; and that customers be provided with the right to access and correct their personal data. The proposed system's customer management module is designed in full compliance with these requirements, with privacy notice integration in the customer enrollment interface, data minimization in the fields collected, role-based access controls limiting customer data access to authorized staff, and secure encrypted storage of all customer records.

Kumar and Reinartz (2012) documented the evolution of CRM systems from simple customer databases to analytically sophisticated platforms capable of predicting customer lifetime value, identifying churn risk, and triggering automated retention interventions. While the proposed system's customer management module is intentionally scoped to the simpler end of this spectrum — customer profiles and purchase history rather than predictive analytics — the data architecture is designed to support more sophisticated CRM capabilities in future development iterations, ensuring that the investment in customer data collection from day one compounds in value as the system and the business's analytical capacity mature.

DIGITAL PROMOTIONS AND COUPON MANAGEMENT

The management of promotional campaigns — discount offers, limited-time specials, bundled meal deals, and loyalty rewards — represents a significant operational and marketing function for multi-branch food service businesses. In the current manual environment at Luna's Arrozcaldo, promotional offers are communicated to cashier staff through briefings and printed notices, and tracking the redemption and effectiveness of promotions is practically impossible without significant manual data collection effort. The proposed system's promotions management module transforms this function by enabling the creation of digital promotional offers with defined parameters — discount amounts, validity periods, applicable products and branches, redemption limits — and automatically tracking redemption rates and revenue impact through the transaction database.

Neslin et al. (2006) conducted a comprehensive review of promotional management research in the retail and food service sectors, finding that businesses with systematic, data-driven promotional management consistently outperformed those with ad-hoc promotional practices on key metrics including average transaction value, visit frequency, and customer acquisition costs. The proposed system's analytical reporting module includes promotional performance dashboards — redemption rates, revenue impact per promotion, branch-level redemption distributions — that provide exactly the data-driven visibility that Neslin et al. identified as the distinguishing characteristic of effective promotional management.

The coupon and promotions management module also provides an important internal control function: by defining promotional offers at the system level rather than leaving discount application to the discretion of individual cashiers, the system ensures that discounts are applied consistently, correctly, and only to eligible transactions. This consistency both protects revenue integrity and provides customers with a predictable, fair promotional experience regardless of which branch or cashier they interact with.

ROLE-BASED ACCESS CONTROL AND INFORMATION SECURITY

The security architecture of any business information system handling financial transaction data, customer personal information, and operational performance data must be designed with care to protect sensitive information from unauthorized access and to ensure compliance with applicable legal requirements. For the proposed system, the primary security architecture mechanism is role-based access control (RBAC), which restricts each user's system access to the features and data appropriate to their organizational role.

Ferraiolo and Kuhn (1992) formulated the foundational RBAC model, defining roles as collections of permissions that can be assigned to users rather than assigning permissions directly to individual user accounts. This role-centric approach simplifies administration in organizations where staff membership in roles changes more frequently than the role definitions themselves — a characteristic that accurately describes Luna's Arrozcaldo, where branch cashier staff turnover is relatively high but the cashier role's required system permissions remain constant. When a new cashier joins a branch, their user account is assigned the cashier role and immediately receives the appropriate set of permissions; no individual permission configuration is required.

The three-role RBAC model implemented in the proposed system — Cashier, Branch Manager, and Administrator — reflects a careful analysis of the operational roles and information access requirements of Luna's Arrozcaldo's three primary user groups. Cashiers require access to the POS terminal and the ability to view current inventory levels at their branch, but should not be able to view network-wide sales reports, modify product prices, or access customer data. Branch Managers require access to their branch's sales reports, inventory management tools, and staff activity logs, but should not be able to access data from other branches or modify system-wide settings. Administrators — the business owner and any designated management staff — require full access to all system functions and data across the entire branch network.

The implementation of password security in the proposed system uses PHP's bcrypt hashing function with an appropriate work factor, following current best practices for credential storage in web applications. Bcrypt is specifically designed for password hashing: it is computationally expensive to calculate, making brute-force attacks infeasible; it incorporates a random salt value that prevents precomputed dictionary attacks; and its work factor can be increased over time as computing power increases, maintaining security strength as the threat environment evolves. Van Oorschot (2021) identified bcrypt as among the most appropriate choices for password hashing in web applications, particularly those accessible to users without specialized security knowledge.

Transport Layer Security (TLS) encryption of all data transmission between the client browser and the Supabase-hosted backend is a non-negotiable security requirement for any system handling financial transaction and personal customer data. TLS ensures that transaction data, customer information, and management credentials cannot be intercepted and read by unauthorized parties on the network path between the Luna's Arrozcaldo branch workstations and the cloud database. Supabase enforces TLS for all database connections and API calls, providing this protection as a built-in platform feature.

The complete audit trail maintained by the proposed system — recording the timestamp and user ID for every transaction, inventory adjustment, login event, and administrative action — serves both security and operational management functions. From a security perspective, the audit trail enables the detection of unauthorized or anomalous activity: if a transaction is reversed or a price is modified, the audit trail records exactly who performed the action and when. From an operational management perspective, the audit trail provides the data necessary to identify systematic cashier errors, investigate inventory discrepancies, and hold staff accountable for operational compliance. Gollmann (2011) identified comprehensive audit logging as one of the most valuable security controls available to organizations operating information systems with multiple users of varying trust levels — a description that precisely fits the Luna's Arrozcaldo deployment context.

CULTURAL ECONOMICS AND HERITAGE FOOD BUSINESSES

The intersection of cultural identity, community heritage, and digital technology presents distinctive challenges and opportunities not fully addressed by the mainstream information systems literature, which has tended to focus on large, commercially oriented enterprises with limited cultural specificity. For food businesses like Luna's Arrozcaldo — institutions whose value proposition is inseparable from their cultural authenticity and community embeddedness — the design and implementation of digital systems requires a sensitivity to cultural context that goes beyond technical competence.

The cultural economist David Throsby (2001) articulated the concept of cultural capital as the store of cultural value embedded in cultural assets — whether tangible (heritage buildings, artworks) or intangible (traditional practices, community knowledge, brand identity). Applying this framework to food businesses, Luna's Arrozcaldo's most valuable asset is not its kitchen equipment, its branch locations, or its supply chain — it is the 57-year accumulation of community trust, cultural authenticity, and Ilonggo food heritage that its brand represents. This cultural capital is simultaneously the business's greatest competitive advantage and its most fragile asset: it can be inadvertently undermined by changes that signal a departure from the values — accessibility, community service, genuine Ilonggo character — that customers associate with the institution.

This cultural sensitivity has important implications for digital system design. A POS interface that feels cold, corporate, or alien to the business's warm service character could subtly undermine the experience that customers associate with Luna's hospitality. The proposed system's interface has been designed with this concern explicitly in mind: the visual design uses warm tones consistent with the brand's food heritage identity, menu item names and categories are presented in the familiar language of Luna's menu boards, and the system's operational logic accommodates the informal service customs — the generous portion adjustments, the familiar customer recognition — that characterize the Luna's Arrozcaldo experience.

Iloilo City's designation as a UNESCO Creative City of Gastronomy in 2024 has created a new context for understanding the significance of heritage food institutions like Luna's Arrozcaldo within the city's economic and cultural landscape. The UNESCO Creative Cities of Gastronomy designation recognizes cities that have built their creative economies and cultural identities around food traditions, and comes with a commitment to using food as a medium for sustainable urban development, social cohesion, and cultural preservation. In this framework, Luna's Arrozcaldo is not merely a private business but a cultural institution whose continued health contributes to the city's fulfillment of its international commitments to food heritage preservation.

The tension between tradition and technology in heritage food businesses has been explored by food scholars including Cwiertka (2006), who documented how Japanese traditional food establishments have navigated the adoption of modern management technologies while preserving the craft identity and cultural authenticity that define their value. Her analysis identified several strategies that successful heritage food businesses have used to reconcile tradition and technology: containing technology to back-of-house and management functions, keeping customer-facing service interactions as warm and personal as possible, framing technology adoption as a means of protecting rather than replacing traditional values, and involving long-term staff in technology design decisions. All four of these strategies have been applied in the design and implementation of the proposed system for Luna's Arrozcaldo.

THE PHILIPPINE FOOD HERITAGE AND ILONGGO CULINARY IDENTITY

The Philippines' rich and regionally diverse food heritage has been documented by food historians including Doreen Fernandez, whose foundational work Tikim (1994) established the academic framework for understanding Philippine food culture as a living heritage constantly shaped by the intersection of indigenous Austronesian traditions, colonial influences, and contemporary innovation. The Ilonggo culinary tradition that Luna's Arrozcaldo embodies represents one of the most distinctive and celebrated strands of this broader Philippine food heritage — characterized by a sophisticated balance of flavors, a strong emphasis on freshness and quality ingredients, and a deeply embedded social function as the food of community celebration and everyday nourishment alike.

Tupas and Lorente (2018) documented the economic and cultural significance of heritage food businesses in Philippine regional cities, finding that such businesses serve simultaneously as commercial enterprises, community gathering places, and repositories of regional cultural knowledge. Their research found that heritage food businesses that successfully adopted digital management tools without compromising their cultural authenticity consistently outperformed those that either rejected digital adoption entirely or adopted digital tools in ways that disrupted their community-oriented service character. The proposed system is designed in direct alignment with this finding: digital efficiency in the back-office and management layer, cultural warmth and community orientation preserved in the customer-facing service experience.

Iloilo City's food culture has been documented as among the most sophisticated in the Philippines, reflecting the city's layered history as a center of trade, education, and civil society in the Western Visayas region. Chua (2019) traced the distinctive character of Ilonggo cuisine — its emphasis on freshness, its restrained use of fat and strong seasoning relative to other Philippine regional styles, its sophisticated use of native ingredients — to the cultural values of the Ilonggo people: a commitment to quality, a preference for subtlety over excess, and a deep pride in local heritage. These cultural values are embodied in the food that Luna's Arrozcaldo has served since 1967, and they have informed the research team's understanding of what it means to develop technology that is truly appropriate for this specific institutional and cultural context.

The specific challenges of digital transformation for food businesses in the Western Visayas region have received limited direct attention in the academic literature, reflecting the broader under-representation of regional Philippine perspectives in information systems research. This study aims to contribute to filling that gap by documenting, in rigorous and accessible detail, the specific design and implementation decisions required to develop an appropriate digital system for a heritage food MSME in the Iloilo context. The researchers hope that this documentation will be useful not only as a guide for Luna's Arrozcaldo's continued digital development but as a reference for future researchers and practitioners working with similar businesses throughout the Visayas region and the broader Philippine provincial landscape.

RELATED SYSTEMS AND LOCAL STUDIES

Several studies conducted by Philippine researchers at various institutions have documented the development and evaluation of digital systems for food service businesses that share significant characteristics with the proposed system for Luna's Arrozcaldo. These related systems provide both a benchmark against which the proposed system's design can be evaluated and a source of practical lessons learned from comparable implementation contexts.

Encarnacion and Genodia (2016) documented the development of an online ordering system for H&J Bakeshop, a single-location bakery business in Iloilo City. Their study identified ease of use and order accuracy as the primary success metrics for a food service digital system and found that the most significant barrier to staff adoption was the initial learning curve associated with abandoning familiar manual processes. Their recommendation — that training should be embedded in the system interface itself rather than conducted as a separate pre-deployment exercise — has been incorporated into the proposed system's design through contextual help text, guided workflows, and real-time error messages that coach users through correct procedures.

The Bugana electronic marketplace study (Barredo et al., 2023) represents the most directly comparable Philippine study to the proposed research, documenting a digital transaction system implemented for a food producer in Negros Occidental. The study's key finding — that digital transaction recording reduced reconciliation time by 40 percent and essentially eliminated arithmetic errors in sales totals — provides strong empirical support for the anticipated operational benefits of the proposed Luna's Arrozcaldo system. The Bugana study's documentation of implementation challenges, particularly the initial resistance of older staff members to digital transaction entry, has informed the proposed system's training and change management approach.

The PIDS study by Briones et al. (2023) on transforming Philippine agri-food systems with digital technology provides the broadest national policy context for the proposed research. Their analysis found that the most impactful interventions were those addressing the entire supply chain from producer to consumer, rather than optimizing a single point. While the proposed system focuses primarily on the internal sales and management functions of Luna's Arrozcaldo, the PIDS framework highlights the future potential for extending the system's scope to include supplier relationship management and procurement tracking in subsequent development phases.

Gañgan's (2025) study of the Agri-Pinoy Trading Center web-based market goods trading system provides the most recent Philippine case study of a web-based transaction system for a food-sector business. The study's documentation of the PostgreSQL database design choices — particularly the normalization strategy for product, transaction, and user tables — closely parallels the database architecture of the proposed system and provides valuable validation that the proposed schema design is appropriate and robust for the scale and operational context of the Luna's Arrozcaldo deployment.

Beyond the specific studies reviewed above, the research team surveyed a broader landscape of Philippine institutional capstone research on digital systems for food businesses, finding a consistent pattern: systems developed with active client participation throughout the development process consistently outperformed those developed from upfront specifications with limited client involvement. This finding, consistent across diverse institutional and geographic contexts within the Philippines, provides additional support for the Agile co-design methodology employed in this study and reinforces the importance of the Sprint review sessions with Luna's Arrozcaldo staff as a central pillar of the research design.

PHP AND WEB-BASED TECHNOLOGIES IN BUSINESS INFORMATION SYSTEMS

PHP (Hypertext Preprocessor) is a widely-used, open-source, server-side scripting language that has been the foundational technology of web-based business applications since the mid-1990s. Its dominance in the web application development ecosystem — particularly among small-to-medium development teams with constrained budgets and timelines — stems from several characteristics that make it particularly well-suited to the development context of this study: an extensive, well-documented standard library for common web application tasks (session management, form handling, database connectivity, and email generation); a large community of Filipino developers who have built significant expertise with the language through university curricula, bootcamps, and industry experience; and a wide range of hosting providers offering PHP-compatible environments at price points accessible to MSME technology budgets.

Lerdorf (1995) originally developed PHP as a simple tool for tracking visits to his personal website; the language has since evolved through eight major versions, with PHP 8.x representing the current production standard. The PHP 8.x language features used in the proposed system — named arguments, match expressions, typed properties, and union types — significantly improve code readability and maintainability compared to earlier PHP versions, reducing the likelihood of runtime errors and the cost of future maintenance. The research team's choice of PHP 8.x for the proposed system reflects both the current state of the art in PHP development and the availability of developer expertise with this version in the Iloilo City IT community.

The Bootstrap front-end framework, used in the proposed system's user interface, has become the most widely adopted CSS framework for web application development globally, with comprehensive documentation and a large community of practitioners. Bootstrap's grid system provides the responsive layout foundation that allows the management dashboard to render correctly across the range of screen sizes — from the counter-mounted desktop monitors at branch POS terminals to the laptop screen the owner uses for remote management access. Its pre-built component library — form elements, navigation bars, tables, modal dialogs, and alert components — allows the research team to deliver a consistent, professional-quality user interface without the time investment of building custom CSS from scratch, concentrating development effort on the application logic specific to Luna's Arrozcaldo's operational requirements.

The JavaScript ecosystem integrated into the proposed system's frontend includes Chart.js for data visualization in the analytical reporting module. Chart.js provides a lightweight, well-documented library for generating interactive charts — bar charts, line charts, doughnut charts — from structured data, with smooth animations and interactive tooltips that enhance the readability of the management dashboard's performance metrics. Nath and Bhatt (2016) identified data visualization quality as a key determinant of management dashboard usability, finding that interactive, visually clear charts significantly improved the speed and accuracy with which non-specialist users could draw correct conclusions from business performance data. Chart.js's quality and ease of implementation make it the appropriate choice for the proposed system's reporting module.

Docker containerization, used in the proposed system's deployment architecture, provides a significant operational advantage for an MSME system that will be maintained by a development team without dedicated DevOps resources. By packaging the application and all its dependencies into a single, self-contained container image, Docker ensures that the system runs identically in development, testing, and production environments, eliminating the class of environment-specific bugs that commonly plague web application deployments. Merkel (2014) documented Docker's transformative impact on web application deployment practices, noting that containerization reduces deployment time, improves environment consistency, and simplifies the process of rolling back to a previous system version in the event of a deployment problem — all characteristics of particular value in the MSME context where development and operations responsibilities are often shared by the same small team.

HUMAN-COMPUTER INTERACTION IN FOOD SERVICE SYSTEMS

Human-Computer Interaction (HCI) as a discipline is concerned with the design and evaluation of computing systems for human use, with particular attention to the match between system design and the cognitive, physical, and contextual characteristics of the users who will operate the system. For the Luna's Arrozcaldo Digital Sales Processing System, HCI principles are not decorative additions to the system design but foundational requirements: the system must be usable by cashier staff operating under time pressure in a noisy, demanding counter service environment, often with limited prior experience with business software.

Norman (2013), in his foundational text The Design of Everyday Things, articulated the concept of affordances as the perceived and actual properties of objects that signal how they should be used. Applied to user interface design, affordances are the visual cues — button shapes, icon choices, color coding, layout conventions — that tell users what an interactive element does and how to interact with it. The proposed system's POS interface has been designed with explicit attention to affordances: the add-to-order button is large, prominently placed, and uses a universally recognized plus icon; the checkout button is visually distinct from all other interface elements and changes color when an order is ready for payment; error states are displayed in red with explicit text explanations rather than coded error numbers.

Nielsen's (1994) heuristics for user interface design — a set of ten principles derived from usability research across diverse interface types — have been applied as a design checklist throughout the development of the proposed system. The most relevant heuristics for the Luna's Arrozcaldo context are: visibility of system status (the POS terminal always displays the current order total prominently), error prevention (required fields are validated before form submission; potentially destructive actions require confirmation), recognition rather than recall (menu items are displayed as a visual grid rather than requiring cashiers to memorize and type product codes), and help and documentation (contextual help text is available at every interface screen without requiring navigation away from the current task).

Shneiderman's (2016) Eight Golden Rules of Interface Design — which include minimizing short-term memory load, supporting internal locus of control, reducing cognitive load, and designing for consistency — have also informed the proposed system's interface. The consistency principle is particularly important in the multi-branch deployment context: if cashiers from different branches interact with the same interface, the consistency of that interface across all branches reduces training costs and helps staff who transfer between branches adapt quickly without retraining. The proposed system maintains complete interface consistency across all branch deployments, with branch-specific content (branch name, inventory data) surfaced through the data layer rather than interface customization.

THE PHILIPPINE DIGITAL COMMERCE AND CONNECTIVITY LANDSCAPE

The broader Philippine digital commerce and connectivity landscape provides an important contextual backdrop for understanding the opportunities and constraints that shape the proposed system's design and implementation. The Philippines has experienced dramatic growth in internet penetration and mobile device adoption over the past decade, driven by competitive mobile data pricing, the proliferation of affordable Android smartphones, and the expansion of telecommunications infrastructure into provincial cities like Iloilo. According to the Philippine Statistics Authority's 2023 Survey on Information and Communications Technology (ICT) Access and Use by Households and Individuals, internet penetration among Filipino households had reached 67 percent by 2023, with penetration rates in urban areas like Iloilo City significantly higher than the national average.

This expanded connectivity infrastructure is a prerequisite for the proposed system's cloud-hosted, real-time data architecture. A system that depends on reliable internet connectivity to synchronize branch POS transactions with a central cloud database is only feasible where that connectivity can be reliably provided. The research team's assessment of internet service availability at each of Luna's Arrozcaldo's eight branch locations confirmed that fiber-based or LTE broadband connections of sufficient reliability and bandwidth for the proposed system's requirements are available at all eight locations, with backup mobile data connectivity as a failover option for continuity of POS operations during temporary primary connection outages.

The Philippine government's commitment to digital infrastructure development, articulated in the Philippine Digital Infrastructure Project and the National Broadband Plan, provides a positive long-term context for MSME digital transformation investments. As connectivity infrastructure continues to improve across Iloilo City and the Western Visayas region, the proportion of potential MSME customers who can be served through digital channels will continue to expand, creating ongoing incentives for food service businesses to invest in digital systems. The proposed system's web-based architecture positions Luna's Arrozcaldo to benefit from this expanding digital infrastructure without requiring costly system replacement as connectivity conditions improve.

The rapid growth of digital payment adoption in the Philippines — driven by GCash, Maya, and bank-linked QR payment systems — has created both an opportunity and an expectation for food service businesses. Customers who regularly pay for other purchases digitally increasingly expect the option to do so at their favorite food establishments. While the proposed system's initial scope does not include direct integration with digital payment platforms — this integration is identified as a future development priority — the system's modular architecture is explicitly designed to accommodate this integration in subsequent development phases without requiring structural changes to the database or core application logic.

The broader expansion of the Philippine digital economy has been accompanied by growing governmental attention to data governance and cybersecurity at the MSME level. The Department of Information and Communications Technology's guidelines for MSME information security, aligned with international standards including ISO/IEC 27001, emphasize that even small businesses handling customer personal data are subject to data protection obligations and should implement baseline security controls proportionate to the sensitivity of the data they handle. The proposed system's security architecture — role-based access control, encrypted data transmission, bcrypt password storage, and comprehensive audit logging — reflects these guidelines and ensures that Luna's Arrozcaldo's digital transformation proceeds on a foundation of responsible data governance from the outset.

SYNTHESIS AND RESEARCH GAP

The review of related literature conducted for this study reveals a substantial and growing body of research on digital transaction systems, business intelligence tools, inventory management platforms, customer loyalty systems, and digital transformation frameworks — all of which are directly relevant to the proposed system for Luna's Arrozcaldo. This literature provides both the theoretical grounding for the system's design and empirical evidence for the operational benefits its implementation is anticipated to produce.

The seven thematic areas reviewed in this chapter together establish several key propositions that have guided the proposed system's design. First, that web-based POS systems can meaningfully reduce transaction processing time and eliminate arithmetic errors in food service environments with mixed-literacy staff, when the interface is designed with the specific cognitive and environmental demands of counter service work in mind. Second, that Agile methodology produces superior outcomes for MSME digital system development when client requirements are complex, partially specified, or likely to evolve during development — all of which characterize the Luna's Arrozcaldo context. Third, that centralized data architecture and business intelligence capabilities create management value that extends far beyond the operational efficiency gains of individual digital modules. Fourth, that real-time inventory management with threshold alerts can substantially reduce both stockout incidents and over-ordering waste in multi-branch food service businesses. Fifth, that digital CRM and loyalty program capabilities create measurable improvements in customer visit frequency when enrollment processes are streamlined and privacy compliance is maintained. Sixth, that RBAC security architecture and comprehensive audit logging provide both data protection and operational accountability in multi-user business information systems. And seventh, that the cultural authenticity and community identity of heritage food businesses must be actively protected — through deliberate interface design choices and participatory design processes — when digital systems are introduced.

The review also reveals a significant research gap: while individual components of the proposed system have been extensively studied in isolation, there are very few studies documenting the development and evaluation of integrated systems combining all of these components for a single multi-branch MSME food business in the Philippine regional context. Most closely comparable studies either focus on a single functional domain or are conducted in Metro Manila, with limited attention to the specific operational and cultural dynamics of regional city food businesses like those in Iloilo.

This study addresses the identified gap directly by documenting the end-to-end design, development, and evaluation of a comprehensive, integrated digital system that simultaneously addresses all seven operational challenges identified in Chapter I in a single coherent platform, tailored to the specific operational context, cultural identity, and resource constraints of a heritage food MSME in Iloilo City. Furthermore, this study makes a methodological contribution to the Philippine MSME digital transformation literature by demonstrating the value of an Agile development approach in which real end users participate actively in the design and testing of the system throughout the development process, rather than serving as passive subjects of a pre-defined requirements specification.

The convergence of findings across all seven thematic areas reviewed in this chapter affirms the central proposition of this study: that the careful, culturally sensitive, and user-centered design of a digital information system for a heritage food MSME can produce not only technical functionality but genuine organizational value — improving efficiency, reducing errors, enhancing management visibility, and strengthening community relationships — while fully preserving the cultural authenticity and community spirit that define the institution's identity and sustain its competitive advantage for generations to come.

IMPLICATIONS FOR SYSTEM DESIGN AND FUTURE RESEARCH

The literature reviewed in this chapter carries several direct implications for the architectural and design decisions embedded in the proposed system. The convergence of findings across POS research, MSME digital transformation studies, and HCI literature points toward a consistent design philosophy: that the most successful digital systems for MSME food service businesses are those that prioritize simplicity and reliability over feature complexity, that embed training and guidance within the interface rather than requiring separate orientation programs, and that are introduced in close collaboration with the staff who will operate them daily.

The evidence on Agile methodology's superior performance in MSME digital system development contexts has shaped not only the process by which the proposed system was built, but the architecture of the system itself. An Agile-aligned architecture is modular and extensible — built in components that can be independently developed, tested, and deployed — rather than monolithic. This modularity has allowed the research team to prioritize the POS and inventory modules for the initial deployment while laying the architectural groundwork for the CRM, promotions, and advanced analytics modules that will be developed in subsequent phases.

The literature on cultural economics and heritage food institutions reinforces a point that the research team has internalized throughout this project: that the purpose of a digital system is always to serve the human institution it supports, never to displace or diminish it. Luna's Arrozcaldo has earned its place in Iloilo City's culinary landscape through 57 years of consistent quality, genuine community care, and the stewardship of a family whose values are inseparable from the food they serve. The proposed system is offered in that spirit — as a tool that frees the institution's people to focus more fully on what they do best, by removing the burden of manual processes that have been consuming time and energy that could be better spent on the food, the customers, and the community.

Future research building on this study could productively explore several directions. The longitudinal evaluation of the proposed system's operational impact — measuring actual reductions in reconciliation time, stockout incidents, and management travel burden in the months following deployment — would provide empirically grounded evidence for the operational benefits that the literature predicts and this study's design anticipates. A comparative study examining digital transformation outcomes across multiple heritage food

MSMEs in Western Visayas would extend the generalizability of this study's findings beyond the single-case context. And the development of culturally adapted digital transformation frameworks specifically for Philippine provincial food businesses would contribute a much-needed regional perspective to a literature currently dominated by Metro Manila and international contexts.

The integration of artificial intelligence and machine learning capabilities into MSME business information systems represents a frontier that is rapidly approaching accessibility for businesses at Luna's Arrozcaldo's scale. Demand forecasting models trained on the historical transaction data that the proposed system will generate, natural language interfaces that allow non-technical staff to query the database in conversational Filipino or Hiligaynon, and computer vision systems for automated inventory counting are among the AI-enabled capabilities that may become practical implementation options within the next three to five years. The proposed system's data architecture has been designed with this future in mind: the transaction, inventory, and customer data structures are normalized and comprehensively logged in ways that will support machine learning model training when that capability becomes accessible.

The proposed system also has implications for how PHINMA University of Iloilo's College of Information Technology Education positions its capstone research program within the regional information systems ecosystem. By demonstrating that student research teams can develop genuinely production-ready digital systems for heritage MSME clients — systems that address real operational problems with real technical solutions, grounded in relevant academic literature and validated through rigorous user testing — this study models a form of community-engaged scholarship that creates value simultaneously for the institution, the client organization, and the academic field. The researchers hope that this study will serve as a reference point for future capstone teams undertaking comparable projects, and as evidence of the genuine contribution that university-based applied research can make to the digital transformation of Iloilo City's MSME community.

REFERENCES

Alampay, E., Villaroman, N., & Santos, J. (2019). Digital loyalty programs in Philippine casual dining: Adoption patterns and impact on visit frequency. Philippine Management Review, 26(1), 45–62.

Andal-Ancion, A., Cartwright, P. A., & Yip, G. S. (2020). Digital transformation in Southeast Asian MSMEs during COVID-19: Acceleration and implications. Journal of International Business Studies, 51(8), 1203–1219.

Barredo, L. D., Limen, J. R., Galve, C. M., Dewara, J. M., & Bucao, W. F. (2023). Bugana: Electronic business to consumer marketplace for farmers' products of Bago City. Kabatiran ICT Journal, 1(1), 69–74. https://doi.org/10.61864/kabatiran.v1i1.43

Bhatt, A. K., & Bhatt, R. (2012). Food waste reduction through digital inventory management in multi-unit food service operations. Journal of Foodservice Business Research, 15(3), 221–238.

Briones, R. M., Galang, I. M. R., & Latigar, J. S. (2023). Transforming Philippine agri-food systems with digital technology: Extent, prospects, and inclusiveness. Philippine Institute for Development Studies Discussion Paper 2023-29. https://doi.org/10.62986/dp2023.29

Chaudhuri, S., Dayal, U., & Narasayya, V. (2011). An overview of business intelligence technology. Communications of the ACM, 54(8), 88–98.

Chopra, S., & Meindl, P. (2016). Supply chain management: Strategy, planning, and operation (6th ed.). Pearson.

Chua, J. (2019). The flavors of Iloilo: A culinary and cultural history of Ilonggo cuisine. University of the Philippines Press.

Cwiertka, K. J. (2006). Modern Japanese cuisine: Food, power and national identity. Reaktion Books.

Date, C. J. (2003). An introduction to database systems (8th ed.). Addison-Wesley.

Davis, F. D. (1989). Perceived usefulness, perceived ease of use, and user acceptance of information technology. MIS Quarterly, 13(3), 319–340.

Davis, J., & Hamilton, S. (2016). Inventory alert systems and stockout reduction in quick-service restaurants. Journal of Foodservice Business Research, 19(4), 312–328.

De Jesus, R. L. (2017). E-commerce and e-marketplace usage in selected small and medium agribusiness food manufacturing enterprises in Cavite (Undergraduate thesis). University of the Philippines Los Baños.

Dingsøyr, T., Nerur, S., Balijepally, V., & Moe, N. B. (2012). A decade of agile methodologies: Towards explaining agile software development. Journal of Systems and Software, 85(6), 1213–1221.

Encarnacion, C., & Genodia, J. (2016). Online ordering system for H&J Bakeshop. Capstone Research, PHINMA University of Iloilo.

Fernandez, D. G. (1994). Tikim: Essays on Philippine food and culture. Anvil Publishing.

Ferraiolo, D., & Kuhn, R. (1992). Role-based access control. Proceedings of the 15th National Computer Security Conference, 554–563.

Few, S., & Edge, P. (2006). Dashboard design: Beyond meters, gauges, and traffic lights. Business Intelligence Journal, 11(1), 20–29.

Gañgan, M. G. T. (2025). Web-based market goods trading system of Agri-Pinoy Trading Center. International Journal of Multidisciplinary Research and Arts, 8(6). https://doi.org/10.47191/ijmra/v8-i06-59

Gollmann, D. (2011). Computer security (3rd ed.). Wiley.

Highsmith, J., & Cockburn, A. (2001). Agile software development: The business of innovation. Computer, 34(9), 120–127.

Kassim, N. M., & Baharudin, Z. (2017). Participatory POS design and adoption outcomes in Southeast Asian quick-service restaurants. International Journal of Hospitality Management, 62, 88–97.

Kimes, S. E. (2011). The current state of online food ordering in the U.S. restaurant industry. Cornell Hospitality Report, 11(17), 6–18.

Kivela, J., Inbakaran, R., & Reece, J. (2000). Consumer research in the restaurant environment: Part 3 — Analysis, findings and conclusions. International Journal of Contemporary Hospitality Management, 12(1), 13–30.

Kumar, V., & Reinartz, W. (2012). Customer relationship management: Concept, strategy, and tools (2nd ed.). Springer.

Momjian, B. (2001). PostgreSQL: Introduction and concepts. Addison-Wesley.

Muller, C. C. (2019). Restaurant management: Customers, operations, and employees (3rd ed.). Pearson.

Negash, S. (2004). Business intelligence. Communications of the Association for Information Systems, 13(1), 177–195.

Neslin, S. A., Grewal, D., Leghorn, R., Shankar, V., Teerling, M. L., Thomas, J. S., & Verhoef, P. C. (2006). Challenges and opportunities in multichannel customer management. Journal of Service Research, 9(2), 95–112.

Olsen, M. D., & Connolly, D. J. (2000). Experience-based travel: How technology is changing the hospitality industry. Cornell Hotel and Restaurant Administration Quarterly, 41(1), 30–40.

Parfitt, J., Barthel, M., & Macnaughton, S. (2010). Food waste within food supply chains: Quantification and potential for change to 2050. Philosophical Transactions of the Royal Society B, 365(1554), 3065–3081.

Parsa, H. G., & Khan, M. A. (2011). Restaurant operations management: Principles and practices. Wiley.

Peppers, D., & Rogers, M. (1993). The one to one future: Building relationships one customer at a time. Currency Doubleday.

Quimba, F. M. A., & Calizo, S. C., Jr. (2019). Determinants of e-commerce adoption of Philippine businesses. Philippine Institute for Development Studies Discussion Paper 2019-30.

Sharma, S., Sarkar, D., & Gupta, D. (2012). Agile processes and methodologies: A conceptual study. International Journal on Computer Science and Engineering, 4(5), 892–898.

Shin, D. H., & Leem, C. S. (2011). A study on usability of enterprise mobile service: Toward a theoretical framework. Industrial Management & Data Systems, 111(6), 839–854.

Throsby, D. (2001). Economics and culture. Cambridge University Press.

Tupas, E. F., & Lorente, R. C. (2018). Heritage food establishments and digital transition in Philippine regional cities: Challenges and opportunities. Journal of Philippine Development, 45(2), 23–47.

Van Oorschot, P. C. (2021). Computer security and the internet: Tools and jewels (2nd ed.). Springer.

Wailgum, T. (2013). Cloud-based point-of-sale systems for small retailers: Benefits, risks, and recommendations. CIO Magazine, March 2013.

Westerman, G., Bonnet, D., & McAfee, A. (2014). Leading digital: Turning technology into business transformation. Harvard Business Review Press.

Wixom, B. H., & Watson, H. J. (2010). The BI-based organization. International Journal of Business Intelligence Research, 1(1), 13–28.

— End of Chapter II —

Design and Development of a Digital Sales Processing System
for Luna's Arrozcaldo — Ilonggo Legacy Since 1967
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Chapter 3

3.1 Research Design

This study will utilize a developmental research design, which is appropriate for projects that involve the systematic process of designing, creating, and evaluating an information system. The focus of developmental research is not only to describe or analyze a phenomenon but also to produce a functional product—in this case, a Digital Sales Processing System with Centralized Sales Management and Analytical Reporting.

Developmental research is widely used in system development studies because it allows researchers to follow a structured and iterative process. This approach ensures that the system being developed is based on identified needs, is continuously improved through evaluation, and is validated through user feedback. In the context of this study, the design aims to address inefficiencies in traditional or manual sales processing systems by introducing a centralized and automated platform that improves accuracy, accessibility, and reporting capabilities.

The research design will follow a phased approach, typically consisting of analysis, design, development, implementation, and evaluation. In the analysis phase, the researchers will gather information regarding current sales processing practices, identify problems encountered by users, and determine the specific requirements needed for the system. This may involve interviews, observations, or surveys from potential users such as business owners, sales staff, or administrators.

In the design phase, the gathered requirements will be translated into system specifications. This includes designing the system architecture, database structure, user interface layouts, and workflow processes. Special attention will be given to ensuring that the system supports centralized sales management, where all sales data are stored and accessed in a unified platform, and analytical reporting, which enables users to generate meaningful insights from sales data.

The development phase involves the actual coding and construction of the system. The researchers will build the web-based application using appropriate programming languages, frameworks, and database systems. During this phase, features such as user authentication, sales recording, data management, and report generation will be implemented.

After development, the system will undergo the implementation phase, where it will be deployed and tested in a controlled environment. Selected users will interact with the system to perform real or simulated sales transactions to evaluate its functionality and usability.

Finally, the evaluation phase will assess the system's performance based on criteria such as functionality, reliability, usability, and efficiency. Feedback from users will be collected to identify areas for improvement. Necessary revisions will then be made to enhance the overall performance and usability of the system.

Overall, the developmental research design is suitable for this study because it ensures that the final output is not only theoretically sound but also practically useful. It allows continuous refinement of the system, ensuring that the Digital Sales Processing System with Centralized Sales Management and Analytical Reporting effectively addresses the needs of its intended users and improves the overall sales management process.

3.2 System Development methodology

The study will adopt the Agile Development Model as the primary system development methodology for the creation of the Digital Sales Processing System with Centralized Sales Management and Analytical Reporting. Agile is a flexible and iterative approach to software development that emphasizes continuous improvement, user collaboration, and adaptive planning. This methodology is highly suitable for systems that require frequent refinement based on user feedback and evolving requirements, especially in business-related applications such as sales management systems.

Unlike traditional linear models, Agile allows the development process to be broken into smaller, manageable cycles known as iterations or sprints. Each iteration produces a working version of the system that can be reviewed and improved. This ensures that the final output is closely aligned with user needs and expectations. The Agile model also encourages active communication between developers and users, which helps in identifying issues early and implementing solutions efficiently.

In this study, the Agile Development Model will be carried out through five main phases: Requirements Analysis, System Design, Development, Testing, and Deployment (Prototype). Each phase plays a critical role in ensuring the successful development of the system.

- Requirements Analysis

  The first phase of the Agile Development Model is the Requirements Analysis phase. This stage focuses on identifying and understanding the needs of the users and the problems that the system aims to solve. For this study, the researchers will gather information from potential users such as business owners, sales personnel, and administrators who handle sales transactions and reporting.

  Data collection methods such as interviews, observations, and surveys will be used to determine the current challenges in sales processing. Common issues may include manual recording errors, difficulty in tracking sales data, lack of centralized information storage, and limited reporting capabilities. The gathered requirements will serve as the foundation for the system's features and functionalities.

  During this phase, both functional requirements and non-functional requirements will be defined. Functional requirements include system capabilities such as user login, sales entry, inventory tracking, and report generation. Non-functional requirements include system performance, security, usability, and reliability. The outcome of this phase is a clear and well-documented list of system requirements that will guide the entire development process.

- System Design

  The second phase is the System Design phase, where the requirements gathered are translated into a structured system blueprint. This phase focuses on how the system will function and how different components will interact with each other.

  In this study, the system design will include the creation of system architecture, database design, and user interface design. The architecture will define how the frontend, backend, and database layers communicate with each other. The database design will focus on organizing data related to sales transactions, user accounts, product information, and reports in a centralized structure to ensure consistency and efficiency.

  Additionally, wireframes and mockups will be developed to represent the user interface design. This will help visualize how users will interact with the system, ensuring that the interface is user-friendly, intuitive, and easy to navigate. Special attention will be given to designing dashboards that support analytical reporting, allowing users to view summaries, charts, and sales trends effectively.

The output of this phase is a complete system design specification that serves as a guide for the development process.

- Development

The Development phase is where the actual coding and construction of the system take place. Based on the system design, developers will begin building the Digital Sales Processing System with Centralized Sales Management and Analytical Reporting.

The system will be developed using appropriate programming languages, frameworks, and database technologies suitable for web-based applications. During this phase, core functionalities will be implemented, including user authentication, sales transaction processing, product management, and report generation.

One of the key features developed in this system is centralized sales management, which ensures that all sales data are stored in a single database accessible to authorized users. This eliminates data redundancy and improves data consistency. Another important feature is analytical reporting, which allows users to generate real-time reports and visual representations of sales performance, helping businesses make informed decisions.

The Agile approach allows this phase to be iterative, meaning that development will occur in cycles. After each cycle, a working version of the system will be reviewed and improved based on feedback, ensuring continuous enhancement of the system.

- Testing

The Testing phase is a critical stage in the Agile Development Model. It ensures that the system functions correctly, efficiently, and securely before deployment. In this phase, the developed system will undergo various types of testing, including functional testing, usability testing, and system testing.

Functional testing will verify whether each feature of the system works as intended. For example, it will check if sales transactions are correctly recorded, if reports are accurately generated, and if user authentication functions properly. Usability testing will evaluate how easy the system is to use, ensuring that users can navigate the interface without difficulty. System testing will assess the overall performance, including speed, responsiveness, and reliability.

Any errors, bugs, or inconsistencies discovered during testing will be documented and corrected immediately. This iterative feedback process ensures that the system becomes more stable and reliable with each improvement cycle.

- Deployment

The final phase is Deployment, where the system is introduced as a working prototype. In this stage, the system will be installed and made accessible to selected users for actual or simulated use. The purpose of deploying a prototype is to evaluate the system in a real-world environment and gather final feedback from users.

During this phase, users will interact with the system to perform tasks such as recording sales, generating reports, and managing data. Their feedback will be collected to assess the system's effectiveness, usability, and overall performance.

Any necessary improvements identified during this phase will be documented for future enhancement or iteration. Since Agile is an iterative methodology, the deployment phase does not mark the end of development but rather a step toward continuous improvement.

3.3 System Architecture

The system architecture of the proposed Digital Sales Processing System with Centralized Sales Management and Analytical Reporting is designed using a client-server architecture, which is a widely adopted model in modern web-based and mobile-based information systems. This architecture is especially suitable for systems that require centralized data management, real-time processing, multi-user access, and secure handling of business-critical information such as sales records and analytical reports.

The client-server architecture separates the system into distinct components that communicate with each other over a network. The client side is responsible for user interaction, while the server side handles data processing, business logic, and database management. This separation allows the system to be more organized, scalable, and easier to maintain. It also ensures that complex processing tasks are handled centrally, while users are provided with a simple and efficient interface for interaction.

In this study, the system is divided into three major layers: the client layer (frontend), the server layer (backend), and the database layer (centralized data storage). Each layer plays a specific role in the overall functionality of the system, but all layers work together through structured communication using APIs (Application Programming Interfaces).

Client Layer (Frontend System)

The client layer represents the front-facing part of the system where users directly interact with the application. This layer is responsible for providing an interface that allows users to perform tasks such as encoding sales transactions, managing product information, viewing dashboards, and generating reports. Users of the system may include administrators, sales staff, and business owners, each with different access levels and functionalities.

The main goal of the client layer is to ensure that the system is user-friendly, responsive, and accessible. Since users may have varying levels of technical expertise, the interface must be intuitive and easy to navigate. The system will be designed with clear layouts, organized menus, and responsive elements that adapt to different screen sizes, whether accessed through desktop browsers or mobile devices.

The client interface will also serve as the primary tool for displaying analytical reports and dashboards. These dashboards may include charts, graphs, tables, and summary statistics that provide insights into sales performance. For example, users can view daily, weekly, or monthly sales trends, identify top-selling products, and monitor revenue performance in real time. These visual representations help users make faster and more informed business decisions.

From a technical perspective, the client layer may be developed using web technologies such as HTML, CSS, and JavaScript, along with modern frontend frameworks like React, Angular, or Vue.js. For mobile versions of the system, frameworks such as Flutter or React Native may be used to ensure cross-platform compatibility. These technologies allow the system to maintain a consistent user experience across different devices.

The client layer communicates with the server through RESTful APIs. Whenever a user performs an action, such as submitting a sales entry or requesting a report, the client sends a request to the server. The response from the server is then displayed on the interface in real time. This communication process ensures that users always receive updated and accurate information without needing to manually refresh or reload the system.

Server Layer (Backend System)

The server layer serves as the core processing unit of the entire system. It is responsible for handling all requests from the client, executing business logic, processing data, and communicating with the database. This layer ensures that all system operations are performed correctly, securely, and efficiently.

One of the primary functions of the server is request handling and data processing. When the client sends a request, such as adding a new sales transaction or generating a report, the server receives the request and processes it according to the system's business rules. It validates the input data to ensure accuracy, checks for completeness, and prevents invalid or duplicate entries from being stored in the database.

The server is also responsible for executing business logic, which defines how the system behaves in different situations. For example, when a sales transaction is recorded, the server calculates the total amount, applies discounts if applicable, updates inventory levels, and stores the final transaction data in the database. By centralizing business logic in the server, the system ensures consistency across all users and prevents discrepancies in data processing.

Another important function of the server is data aggregation and analytics processing. Since the system includes analytical reporting features, the server processes large volumes of sales data to generate meaningful insights. It may analyze patterns such as sales trends over time, product performance, customer purchasing behavior, and revenue distribution. These analytical processes help transform raw data into useful information that supports business decision-making.

The server also plays a major role in security management. It handles user authentication by verifying login credentials and ensuring that only authorized users can access the system. It also manages authorization by assigning different roles and permissions to users, such as administrator, manager, or sales staff. This ensures that users can only access the features and data relevant to their role.

Additionally, the server is responsible for data protection and encryption. Sensitive information such as user credentials and financial data is secured using encryption techniques during transmission and storage. This reduces the risk of unauthorized access and protects the integrity of the system.

Technologies used in the server layer may include Node.js, PHP, Python (Django or Flask), or Java Spring Boot. These backend technologies provide robust frameworks for handling requests, managing databases, and building scalable applications.

The server communicates with the database layer using structured queries and retrieves or updates data as needed. It then sends processed results back to the client through API responses.

Database Layer (Centralized Data Storage)

The database layer is responsible for storing and managing all system data in a structured and centralized manner. In this system, the database supports the concept of centralized sales management, meaning all users interact with a single source of truth for all sales-related information.

The database stores various types of data, including sales transactions, product inventories, user accounts, customer information, and generated reports. By centralizing this data, the system eliminates redundancy and ensures that all users access the same updated information at all times.

A Relational Database Management System (RDBMS) — specifically PostgreSQL through Supabase — is used for this system. The database will be structured using tables that represent different entities. For example, a sales table may store transaction details, a products table may store inventory data, and a users table may manage login credentials and roles. Relationships between these tables will be established using keys to maintain data integrity.

The database also supports data retrieval and reporting functions. When a user requests a report, the server queries the database to retrieve relevant data. This data is then processed and displayed in a structured format such as charts, graphs, or tables on the client interface.

To ensure data reliability, the database will implement backup and recovery mechanisms. Regular backups will be performed to prevent data loss due to system failures, errors, or unexpected events. In case of failure, recovery procedures will allow the system to restore previous data states efficiently.

The database is also optimized for performance to handle large volumes of transactions efficiently. Indexing techniques, query optimization, and proper database normalization will be applied to ensure fast data retrieval and smooth system performance even under heavy usage.

Communication Flow Between Layers

The interaction between the client, server, and database follows a structured communication flow. This flow ensures that all system processes are executed in an organized and efficient manner.

When a user performs an action in the client interface, such as recording a sale or generating a report, the client sends a request to the server through an API. The server receives the request, processes it based on business rules, and interacts with the database if necessary. After completing the required operations, the server sends a response back to the client, which is then displayed to the user.

For example, in a sales transaction process:

11. The user enters sales details in the client interface.
12. The client sends the data to the server via an API request.
13. The server validates the input and applies business rules.
14. The server stores the transaction in the database.
15. The database confirms successful storage.
16. The server sends a response back to the client.
17. The client displays a confirmation message or updated dashboard.

This structured flow ensures consistency, accuracy, and real-time synchronization of data across the entire system.

Advantages of the Client-Server Architecture

The use of a client-server architecture provides several significant advantages for the proposed system. One of the main advantages is centralized data management, which ensures that all

users access a single, consistent database. This reduces duplication of data and improves accuracy.
Another advantage is scalability, as the system can support an increasing number of users and transactions without major structural changes. This is important for businesses that may grow over time and require more advanced system capabilities.
The architecture also enhances security, since all sensitive data is stored and processed on the server rather than on individual client devices. This reduces the risk of data breaches and unauthorized access.
In addition, the system supports real-time data processing, allowing users to receive immediate updates after performing actions. This is particularly important in sales systems where timely and accurate information is essential.
The architecture also improves system maintainability, as updates and improvements can be made on the server side without affecting the client interface. This reduces downtime and simplifies system upgrades.
Finally, it allows for efficient resource management, as heavy processing tasks are handled by the server, while the client focuses only on user interaction and data presentation.
3.4 Tools and Technologies Used
The development of the **Digital Sales Processing System with Centralized Sales Management and Analytical Reporting** requires the use of modern tools and technologies that support efficient system development, scalability, real-time data handling, and secure deployment. The selection of tools is based on their compatibility, performance, ease of use, and ability to support web-based applications with centralized database management and analytical capabilities.
The system utilizes a combination of **frontend technologies, backend programming language, database management system, cloud-based backend services, and development tools**. Each of these plays a significant role in ensuring that the system functions efficiently and meets its intended objectives.
The main technologies used in this study include:

1. Frontend: HTML, CSS, JavaScript
2. Backend: PHP (with integration of Supabase services for modern backend enhancements)
3. Database: Supabase (PostgreSQL-based cloud database)
4. Hosting/Deployment: Railway
5. Development Tools: Visual Studio Code

Each of these tools and technologies is discussed in detail below.
Frontend Technologies

HTML (HyperText Markup Language)

HTML is the foundational markup language used in the development of the system's user interface. It is responsible for structuring the content of web pages. In the context of the Digital Sales Processing System, HTML is used to create the layout of forms, tables, dashboards, navigation menus, and input fields.
HTML enables the system to define elements such as:

1. Sales entry forms

2. Login and registration pages
3. Product management tables
4. Report display sections
5. Dashboard components

The use of HTML ensures that the system has a clear and organized structure, which is essential for usability and accessibility.

CSS (Cascading Style Sheets)

CSS is used to enhance the visual presentation of the system. While HTML provides structure, CSS is responsible for styling and design. It allows the system to be visually appealing and user-friendly.

In this system, CSS is used to:

1. Design responsive layouts for different screen sizes
2. Apply color schemes and themes
3. Style buttons, forms, and tables
4. Improve readability and spacing
5. Create dashboard layouts for analytics

Responsive design is particularly important because users may access the system using different devices such as desktops, laptops, tablets, or smartphones. CSS ensures that the system adapts properly to all screen sizes.

JavaScript

JavaScript is used to add interactivity and dynamic functionality to the system. It allows the system to respond to user actions without requiring page reloads, improving user experience and performance.

In the proposed system, JavaScript is used for:

1. Form validation (e.g., checking empty fields before submission)
2. Dynamic updates of dashboard data
3. Interactive charts and graphs for sales analytics
4. Real-time user feedback messages
5. API communication with backend services

JavaScript plays a key role in making the system more interactive and responsive, especially in handling real-time sales processing and reporting features.

Backend Technologies

PHP (Hypertext Preprocessor)

PHP is used as the primary backend programming language of the system. It is responsible for handling server-side logic, processing user requests, and interacting with the database.

PHP is widely used in web development due to its simplicity, flexibility, and strong database integration capabilities. In this system, PHP performs the following functions:

1. User authentication (login and registration)
2. Processing sales transactions
3. Managing product inventory data
4. Generating reports
5. Communicating with the database (Supabase/PostgreSQL)

PHP acts as the bridge between the frontend interface and the database. When a user submits a request, PHP processes the request, executes business logic, and returns a response to the frontend.

Integration with Supabase (Backend-as-a-Service Enhancement)

In addition to PHP, the system integrates Supabase, a modern backend-as-a-service (BaaS) platform built on PostgreSQL. Supabase provides additional backend capabilities that enhance the system's scalability, real-time features, and authentication system.

Supabase is used in this study as an advanced backend extension to support modern web application requirements. It complements PHP by handling features that require real-time synchronization and cloud-based infrastructure.

Supabase provides several important functionalities:

1. Real-time Database Update

Supabase enables real-time data synchronization, which is important for sales systems. When a new transaction is added, updates are reflected immediately across all connected users without manual refresh.

2. Authentication Services

Supabase provides built-in authentication features such as email login, password management, and session handling. This improves system security and reduces the need for manually building authentication systems in PHP.

3. API Generation

Supabase automatically generates RESTful APIs based on the database structure, making it easier to connect the frontend and backend.

4. PostgreSQL Database Support

Supabase uses PostgreSQL, which supports advanced queries, better scalability, and improved data integrity.

5. Cloud Storage

Supabase can also store files such as receipts, reports, or exported sales documents securely in the cloud.

The integration of Supabase enhances the system by providing modern backend capabilities that improve performance, scalability, and real-time data handling.

Database Technologies

Supabase (PostgreSQL)

Supabase serves as the database layer for this system, built on PostgreSQL, which is a powerful relational database system that supports advanced queries and real-time data handling.

In this system, Supabase is used for:

1. Real-time analytics data storage
2. Advanced querying for reporting
3. Cloud-based database access
4. Synchronization of live sales data

PostgreSQL supports complex queries and advanced indexing, which improves the performance of analytical reporting features in the system.

Hosting and Deployment Technology

Railway (Cloud Deployment Platform)

Railway is used as the deployment and hosting platform for the system. It is a modern cloud platform that simplifies the process of deploying web applications, backend services, and databases.

Railway is chosen because it provides:

1. Easy deployment of PHP applications
2. Cloud hosting for backend services
3. Database hosting integration
4. Automatic scaling capabilities
5. Continuous deployment support

In this system, Railway is used to host:

1. The PHP backend server
2. API services
3. Database connections (Supabase/PostgreSQL)

Railway allows the system to be accessible online, meaning users can access the sales processing system anytime and anywhere using a browser or mobile device.

One of the advantages of Railway is its ability to automatically manage server infrastructure, reducing the need for manual server configuration. This makes deployment faster and more efficient.

Development Tools

Visual Studio Code (VS Code)

Visual Studio Code is used as the primary development environment for the system. It is a lightweight but powerful code editor that supports multiple programming languages and frameworks.

VS Code is used for:

1. Writing HTML, CSS, and JavaScript code
2. Developing PHP backend scripts
3. Managing API integrations
4. Debugging code errors
5. Organizing project files

VS Code also supports extensions that improve productivity, such as:

1. Live Server (for frontend testing)
2. PHP Intelephense (for PHP development support)
3. Git integration tools
4. API testing extensions

The use of VS Code ensures efficient coding, debugging, and project management throughout the development process.

System Integration of Tools and Technologies
The system is built through the integration of all the mentioned technologies. Each tool plays a specific role, but they all work together to form a complete system.
The workflow is as follows:

1. The user interacts with the frontend (HTML, CSS, JavaScript).
2. The frontend sends requests to the backend (PHP + Supabase APIs).
3. The backend processes data and applies business logic.
4. Data is stored and retrieved from Supabase (PostgreSQL) database.
5. The system is hosted and deployed using Railway cloud platform.
6. Developers build and maintain the system using Visual Studio Code.

This integration ensures that the system is fully functional, scalable, and capable of handling real-time sales processing and analytical reporting.

3.5 Data Gathering Techniques

The success of the development of the Digital Sales Processing System with Centralized Sales Management and Analytical Reporting heavily depends on the accuracy, completeness, and relevance of the data collected during the early stages of the study. Data gathering is a crucial phase in the research process because it provides the foundation for identifying system requirements, understanding user needs, and defining the problems that the system aims to solve. Without proper data collection, the resulting system may fail to address real-world issues or may not fully meet user expectations.
In this study, the primary data gathering technique used is the Interview Method. This method is selected because it allows the researchers to obtain detailed, in-depth, and reliable information directly from the intended users of the system. Interviews are particularly effective for system development studies because they enable direct interaction between the researcher and respondents, allowing clarification of responses and exploration of specific concerns related to sales processing and reporting.

Interview as a Data Gathering Technique
The interview method is a qualitative data collection technique that involves asking structured or semi-structured questions to selected respondents in order to gather relevant information about a particular topic. In the context of this study, interviews are conducted to understand the current sales management processes, identify existing problems, and gather user requirements for the proposed system.
Interviews are considered one of the most effective methods for system development research because they allow respondents to freely express their thoughts, experiences, and challenges. Unlike questionnaires, interviews provide opportunities for follow-up questions, which help the researcher gain deeper insights into specific issues related to sales processing systems.
The interview process in this study focuses on stakeholders who are directly involved in sales operations. These include business owners, sales staff, cashiers, and administrators who manage sales transactions and reporting activities. Their experiences are essential in identifying the limitations of current systems and determining the features required in the proposed digital system.

Types of Interview Used

In this study, a semi-structured interview approach is used. This type of interview combines both structured and unstructured methods, allowing the researcher to prepare a set of guiding questions while still giving flexibility for additional follow-up questions based on the respondent's answers.

Semi-structured interviews are ideal for system development research because they balance consistency and flexibility. The prepared questions ensure that all necessary topics are covered, while the open-ended nature of the interview allows respondents to elaborate on their experiences and provide valuable suggestions that may not have been initially considered.

Interview Preparation Process

Before conducting the interviews, the researchers perform careful preparation to ensure the effectiveness and reliability of the data collected. The preparation stage includes the following steps:

1. Identification of Respondents

The first step is identifying the appropriate respondents for the interview. These respondents are individuals who have direct experience with sales processing and reporting systems. They may include:

1. Business owners who oversee sales operations
2. Cashiers or sales personnel who handle daily transactions
3. Administrators responsible for record-keeping and reporting
4. Staff involved in inventory and product management

These individuals are selected because they possess firsthand knowledge of the problems and limitations of current sales systems.

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2. Development of Interview Questions

The next step involves preparing a set of interview questions that will guide the discussion. These questions are designed to extract relevant information about current systems, challenges faced, and desired improvements.

Typical interview questions may include:

1. How do you currently process sales transactions?
2. What challenges do you encounter in your current sales system?
3. How do you record and manage sales data?
4. Do you experience difficulties in generating sales reports?
5. What features would you like to see in a digital sales system?
6. How do you ensure the accuracy of sales records?
7. What problems do you face in tracking inventory or product sales?

These questions are designed to be open-ended to encourage detailed responses.

3. Scheduling of Interviews

After preparing the questions, the researchers schedule interviews at a convenient time for the respondents. Scheduling is important to ensure that participants are available and willing to provide accurate and thoughtful responses without interruptions. Interviews may be conducted in person, through phone calls, or via online communication platforms depending on accessibility and convenience.

Conducting the Interview

During the actual interview process, the researcher follows a structured yet flexible approach. The interview begins with an introduction, where the purpose of the study is explained to the respondent. This helps establish trust and ensures that the participant understands the importance of their contribution.

The researcher then proceeds with asking the prepared questions. While the questions serve as a guide, the interviewer may ask additional follow-up questions based on the responses given by the participant. This allows for deeper exploration of specific issues, particularly those related to inefficiencies in current sales processing systems.

For example, if a respondent mentions difficulties in generating reports, the researcher may ask further questions such as:

1. What specific difficulties do you encounter when generating reports?
2. How long does it usually take to prepare a sales report?
3. Have you experienced errors in your reports before?

This interactive approach ensures that the data gathered is comprehensive and meaningful.

Recording and Documentation of Data

To ensure accuracy, all responses obtained during the interview are documented properly. This may be done through written notes or audio recording (with the consent of the respondent). Recording the interview ensures that no important information is missed and allows the researcher to review responses later during data analysis.

The collected data is then organized based on themes such as:

1. Current system processes
2. Identified problems and limitations
3. User requirements and suggestions
4. Desired system features

This organization helps in simplifying the analysis process and ensures that all relevant information is considered during system development.

Data Analysis from Interviews

After the interviews are conducted, the next step is data analysis. The responses gathered from participants are carefully reviewed and analyzed to identify common patterns, issues, and requirements.

The analysis process focuses on identifying:

1. Repeated problems mentioned by multiple respondents
2. Inefficiencies in current sales processing methods
3. User expectations for a digital system
4. Features that are considered most important

For example, if multiple respondents mention that manual recording of sales leads to errors, this indicates a strong need for an automated sales processing feature in the system.

Similarly, if users express difficulty in generating reports, this highlights the importance of implementing an automated analytical reporting module within the system.

Importance of Interviews in System Development

The interview method plays a critical role in ensuring that the developed system is aligned with real user needs. One of the main advantages of using interviews is that it provides firsthand information directly from users, which is more reliable than assumptions or secondary data sources.

Interviews also allow the researcher to gain a deeper understanding of the context in which the system will be used. This includes understanding how sales transactions are currently handled, what tools are being used, and what challenges users face in their daily operations.

Another important benefit is that interviews allow flexibility in data collection. Unlike structured surveys, interviews allow the researcher to adjust questions based on the flow of conversation. This leads to richer and more detailed information that is valuable for system design and development.

Ethical Considerations in Data Gathering

During the interview process, ethical considerations are strictly observed. Respondents are informed about the purpose of the study and are assured that their responses will be used solely for academic purposes. Consent is obtained before conducting interviews or recording responses.

Confidentiality is also maintained to protect the identity and personal information of respondents. This ensures that participants feel comfortable and are more likely to provide honest and accurate responses.

Relevance to System Development

The data gathered through interviews directly influences the design and development of the Digital Sales Processing System with Centralized Sales Management and Analytical Reporting. The identified problems and user requirements serve as the basis for defining system features such as:

1. Automated sales processing
2. Centralized data storage
3. Real-time reporting
4. User-friendly interface
5. Secure login and access control
6. Analytical dashboards for decision-making

By incorporating user feedback collected through interviews, the system becomes more practical, efficient, and aligned with real-world business needs.

3.6 Testing Procedure

The testing procedure is a critical phase in the development of the Digital Sales Processing System with Centralized Sales Management and Analytical Reporting, as it ensures that the system functions correctly, efficiently, and reliably before it is deployed for actual use. Testing is essential in identifying errors, bugs, inconsistencies, and performance issues that may affect the overall usability and functionality of the system. Without proper testing, the system may fail to meet user requirements or may produce inaccurate results, especially in sales processing and analytical reporting where data accuracy is highly important.

In this study, the primary focus of the testing procedure is System Testing, which evaluates the complete and integrated system as a whole. System testing is performed after all modules and components have been developed and integrated. It ensures that the interaction between different parts of the system works correctly, including the frontend, backend, database, and external services such as APIs or cloud-based tools like Supabase and Railway.

System testing is designed to simulate real-world usage scenarios to determine whether the system meets the specified requirements. It evaluates the system's functionality, performance, security, usability, and reliability under different conditions. This phase is crucial in ensuring that the Digital Sales Processing System operates smoothly when used in actual business environments.

Purpose of System Testing

The main purpose of system testing is to validate that the entire system works as intended and meets the objectives of the study. Specifically, system testing aims to:

1. Verify that all system features function correctly and according to requirements
2. Ensure that data is processed accurately and consistently
3. Identify and fix system errors and bugs before deployment
4. Evaluate system performance under different loads
5. Confirm that security measures are properly implemented
6. Assess the usability and user experience of the system

Since the system involves sales transactions and analytical reporting, accuracy and reliability are critical. Any errors in computation, data storage, or reporting can significantly affect business decisions. Therefore, system testing plays an important role in ensuring the integrity of the entire system.

Scope of System Testing

System testing covers the entire Digital Sales Processing System, including all integrated modules and components. These include:

1. User authentication module (login and registration system)
2. Sales transaction processing module
3. Product and inventory management module
4. Centralized database system
5. Analytical reporting and dashboard module
6. API communication between frontend and backend
7. Integration with Supabase and Railway deployment environment

Each of these components is tested not only individually but also as part of the complete system to ensure proper interaction and functionality.

Types of System Testing Conducted

To ensure comprehensive evaluation, several types of system testing are conducted. Each type focuses on a specific aspect of system performance and functionality.

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1. Functional Testing

Functional testing is performed to verify that each feature of the system works according to the specified requirements. It ensures that all functionalities behave as expected when users interact with the system.

In the context of this system, functional testing includes:

1. Checking if users can successfully log in and log out
2. Verifying if sales transactions are correctly recorded
3. Ensuring that product data can be added, updated, and deleted
4. Confirming that reports are generated accurately
5. Validating that dashboard data reflects real-time updates

Each function is tested using different input scenarios to ensure reliability and correctness.

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2. Integration Testing

Integration testing focuses on the interaction between different modules of the system. Since the system is composed of multiple components such as frontend, backend, database, and external services, it is important to ensure that they work together seamlessly.
This includes testing:

1. Communication between frontend (HTML, CSS, JavaScript) and backend (PHP)
2. Data exchange between PHP backend and Supabase/PostgreSQL database
3. API integration between frontend and Supabase services
4. Data synchronization between Railway-hosted backend and database systems

The goal is to ensure that data flows correctly between components without errors or delays.

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3. Performance Testing

Performance testing evaluates how the system behaves under different workloads. It ensures that the system remains stable and responsive even when handling multiple users or large volumes of data.
For this system, performance testing includes:

1. Measuring response time when loading dashboards and reports
2. Testing system behavior during multiple simultaneous sales transactions
3. Evaluating database query performance under heavy data load
4. Checking system stability during continuous usage

The system must be able to handle real-world business scenarios where multiple users may be accessing and updating data at the same time.

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4. Security Testing

Security testing is conducted to ensure that the system is protected against unauthorized access and data breaches. Since the system deals with sensitive business data such as sales records and user credentials, security is a critical aspect.
Security testing includes:

1. Verifying user authentication (login security)
2. Testing role-based access control (admin, staff, etc.)
3. Ensuring data encryption during transmission
4. Checking protection against unauthorized database access
5. Validating secure API communication between client and server

This ensures that only authorized users can access specific features and data within the system.

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5. Usability Testing

Usability testing evaluates how easy and user-friendly the system is for end-users. Since the system is intended for business use, it is important that users can easily navigate and operate it without technical difficulties.
Usability testing includes:

1. Checking the clarity of the user interface design
2. Evaluating ease of navigation between modules
3. Testing readability of dashboards and reports
4. Assessing user understanding of system features
5. Gathering user feedback on interface improvements

This helps ensure that the system is accessible even to users with minimal technical experience.

Testing Procedure Steps

The system testing process follows a structured procedure to ensure consistency and accuracy in evaluation.

Step 1: Test Planning

In this stage, the scope, objectives, and test cases are defined. The testing team identifies what needs to be tested and prepares test scenarios based on system requirements. This includes creating test cases for all major system functionalities such as sales processing, reporting, and user authentication.

Step 2: Test Case Design

Test cases are developed to simulate real-world usage of the system. Each test case includes input data, expected results, and actual results. For example:

1. Input: Valid login credentials
2. Expected Result: Successful login and dashboard access

This ensures that every function is properly evaluated.

Step 3: Test Execution

During this phase, the actual testing is performed. The system is run under different conditions, and each test case is executed. Observations are recorded, including whether the system behaves as expected or produces errors.

Step 4: Bug Reporting and Debugging

If errors or issues are discovered during testing, they are documented and reported to the development team. Developers then analyze the issues and apply necessary fixes to improve system performance and reliability.

Step 5: Retesting

After fixes are applied, the system is retested to ensure that the issues have been resolved and that no new errors have been introduced. This cycle continues until the system meets all required standards.

Step 6: Final Evaluation

Once all test cases have been successfully passed, a final evaluation is conducted. This confirms that the system is stable, functional, and ready for deployment.

Tools Used in Testing

Several tools and technologies may be used to support the system testing process, including:

1. Browser developer tools for frontend testing
2. Postman for API testing
3. Supabase Table Editor for database testing
4. Supabase dashboard for real-time data monitoring
5. Railway logs for backend performance monitoring

These tools help ensure that all aspects of the system are thoroughly tested and validated.

Importance of System Testing in the Study

System testing is essential because it ensures that the developed system meets its intended purpose. In a sales processing system, even small errors can lead to incorrect financial records or reporting issues. Therefore, testing helps prevent such problems before the system is deployed.

It also ensures that the integration between different technologies such as PHP, Supabase (PostgreSQL), and Railway works smoothly. Since the system relies on multiple platforms, testing ensures that they function as a unified system.

Furthermore, system testing improves user confidence by ensuring that the system is reliable, secure, and easy to use. It also reduces the risk of system failure after deployment, making it a crucial step in the development process.

3.7 Evaluation Criteria

The evaluation criteria for the Digital Sales Processing System with Centralized Sales Management and Analytical Reporting serve as the standards used to assess the overall quality, effectiveness, and readiness of the developed system. Evaluation is a critical phase in system development because it determines whether the system meets its intended objectives and satisfies the needs of its users. In this study, the system will be evaluated based on four major criteria: Functionality, Usability, Security, and Efficiency. These criteria are commonly used in

software engineering and information system development to ensure that a system is not only operational but also reliable, user-friendly, secure, and performant in real-world usage.
The evaluation process involves assessing how well the system performs its intended tasks, how easy it is for users to interact with it, how secure it is against unauthorized access, and how efficiently it utilizes system resources. Each criterion is carefully analyzed to ensure that the system meets both technical and user requirements.

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1. Functionality

Functionality refers to the degree to which the system performs the tasks it was designed to do. It is the most fundamental evaluation criterion because it determines whether the system is actually useful in achieving its main purpose. For the Digital Sales Processing System, functionality focuses on whether all core features work correctly and consistently according to user requirements.
In this system, functionality includes several key components such as sales processing, user management, product management, and analytical reporting. Each of these components must operate correctly and produce accurate results. For example, when a user inputs a sales transaction, the system must correctly calculate totals, apply any discounts if applicable, and store the transaction in the centralized database without errors. Similarly, when generating reports, the system must retrieve accurate data and present it in a meaningful format such as tables, charts, or summaries.
Functional evaluation also includes verifying that all modules of the system are properly integrated. The frontend, backend, and database must work together seamlessly. If a user performs an action in the interface, the system must correctly process the request on the server side and reflect the changes in the database. Any failure in communication between these components would indicate a functional issue.
Another important aspect of functionality is the correctness of analytical reporting features. The system is expected to generate insights based on sales data, such as total revenue, best-selling products, and sales trends over time. These outputs must be accurate and consistent with stored data. Incorrect calculations or missing data would negatively affect decision-making and reduce the reliability of the system.
Functionality also evaluates whether the system supports different user roles. For instance, administrators should have full access to manage users and system settings, while sales staff may only have access to sales entry and viewing reports. Proper role-based access ensures that users can only perform tasks relevant to their responsibilities.
Overall, functionality ensures that the system is capable of performing all required tasks without errors, missing features, or incorrect outputs. It answers the question: Does the system work as intended?

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2. Usability

Usability refers to how easy and efficient the system is for users to understand, learn, and operate. A system may be fully functional, but if it is difficult to use, it will not be effective in real-world applications. Therefore, usability is a critical evaluation criterion, especially for systems intended for business users who may not have advanced technical knowledge.

In the context of the Digital Sales Processing System, usability focuses on the design of the user interface, navigation structure, accessibility, and overall user experience. The system must be designed in a way that allows users to perform tasks with minimal effort and confusion.

One of the key aspects of usability is interface design clarity. The system should have a clean and organized layout, with clearly labeled buttons, forms, and menus. Users should be able to easily identify where to input sales data, how to access reports, and how to navigate between different modules. Overly complex or cluttered interfaces can lead to confusion and errors.

Another important factor is ease of navigation. Users should be able to move between different sections of the system without difficulty. For example, switching from the sales entry page to the reporting dashboard should be smooth and intuitive. A well-structured navigation menu improves efficiency and reduces the learning curve for new users.

Usability also includes system responsiveness, which refers to how quickly the system responds to user actions. A usable system should provide immediate feedback when users submit forms, click buttons, or request data. Delays in response time can negatively affect user experience and productivity.

Additionally, the system should include user feedback mechanisms, such as success messages, error alerts, and confirmation prompts. These messages help users understand whether their actions were successful or if corrections are needed. For example, when a sales transaction is successfully recorded, the system should display a confirmation message.

Accessibility is another important aspect of usability. The system should be designed to accommodate different users, including those accessing it from various devices such as desktops, laptops, tablets, or mobile phones. A responsive design ensures that the system adapts to different screen sizes and maintains usability across platforms.

Training requirements are also considered in usability evaluation. A highly usable system should require minimal training for users to operate effectively. The goal is to make the system intuitive enough that users can quickly learn how to use it without extensive instruction.

Overall, usability ensures that the system is not only functional but also user-friendly and efficient to operate. It answers the question: Can users easily use the system without difficulty?

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3. Security

Security is one of the most critical evaluation criteria in any information system, especially in systems that handle sensitive data such as sales records, user accounts, and financial information. For the Digital Sales Processing System with Centralized Sales Management and Analytical Reporting, security ensures that data is protected from unauthorized access, manipulation, and loss.

The system implements multiple layers of security to protect both user data and system integrity. One of the primary security measures is user authentication, which verifies the identity of users before granting access to the system. Users must provide valid login credentials such as username and password. Without proper authentication, unauthorized users cannot access system features.

In addition to authentication, the system also uses authorization mechanisms. Authorization determines what actions a user is allowed to perform based on their assigned role. For example, an administrator may have full access to manage users and system settings, while a sales staff member may only be allowed to enter and view sales data. This role-based access control helps prevent unauthorized actions within the system.

Another important security feature is data encryption. Sensitive information such as passwords and financial data is encrypted during transmission and storage. This ensures that even if data is intercepted, it cannot be easily read or manipulated by unauthorized parties.

The system also ensures secure communication between components through HTTPS protocols and API security measures. This protects data being transmitted between the client, server, and database from interception or tampering.

Database security is also a key consideration. The centralized database is protected through access controls, ensuring that only authorized server-side applications can interact with it. Direct access from external sources is restricted to prevent data breaches.

Security testing is conducted to identify potential vulnerabilities such as SQL injection, unauthorized access attempts, and data leaks. Any identified vulnerabilities are addressed during the development and testing phases.

Backup and recovery mechanisms also contribute to system security by ensuring that data can be restored in case of system failure or unexpected data loss. Regular backups help maintain data integrity and availability.

Overall, security ensures that the system is protected against threats and that user data remains safe and confidential. It answers the question: Is the system safe from unauthorized access and data breaches?

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4. Efficiency

Efficiency refers to how well the system uses its resources, such as processing power, memory, and network bandwidth, while delivering optimal performance. A highly efficient system performs tasks quickly and smoothly without unnecessary delays or excessive resource consumption.

For the Digital Sales Processing System, efficiency is measured in terms of system response time, data processing speed, and resource utilization. Since the system handles sales transactions and analytical reporting, it must be capable of processing large amounts of data without slowing down.

One aspect of efficiency is response time, which refers to how quickly the system responds to user actions. For example, when a user submits a sales transaction or requests a report, the system should process the request and return results in a short amount of time. Delays in response time can negatively affect user productivity.

Another aspect is database performance. The system uses a centralized database to store and retrieve data. Efficient database design, including proper indexing and query optimization, ensures that data retrieval is fast and accurate even when dealing with large datasets.

The system also considers server performance optimization. Since the backend handles multiple requests from different users, it must be capable of processing these requests concurrently without significant performance degradation. Technologies such as PHP optimization, API caching, and load balancing may be used to improve performance.

Memory and storage efficiency are also evaluated. The system should be designed to minimize unnecessary resource consumption while still maintaining functionality. Efficient coding practices and optimized database structures help achieve this goal.

Another important aspect of efficiency is scalability, which refers to the system's ability to handle increasing workloads without performance issues. As the number of users or amount of data grows, the system should still maintain stable performance.

Cloud deployment using platforms like Railway also contributes to efficiency by automatically managing server resources and scaling the system based on demand. This ensures that the system remains responsive even during peak usage periods.

Overall, efficiency ensures that the system operates smoothly, quickly, and with minimal resource wastage. It answers the question: Does the system perform tasks quickly and effectively using minimal resources?

Data Flow Diagram (DFD – Level 0)

The Data Flow Diagram (DFD) Level 0 for the Digital Sales Processing System with Centralized Sales Management and Analytical Reporting represents the highest-level view of the system's data flow and overall interaction between external users, the system itself, and the centralized database. It provides a simplified but comprehensive illustration of how data moves within the system without going into deep technical details of internal processes.

At this level, the system is viewed as a single process labeled as the Sales & Management System, which interacts with external entities, primarily the User, and communicates with the Database as the central storage unit. The purpose of this diagram is to show how data is input, processed, stored, and retrieved within the system in order to support sales operations, inventory management, user management, and analytical reporting.

The DFD Level 0 is essential in system analysis because it gives a clear overview of the entire system structure in a simple format. It helps developers, researchers, and stakeholders understand how the system functions at a macro level before breaking it down into more detailed processes such as Level 1 and Level 2 DFDs.

Overview of the System Flow

In the proposed system, the flow of data begins with the User, who interacts with the system based on assigned roles such as staff or administrator (admin). Each user role has specific access rights and responsibilities within the system.

The Sales & Management System acts as the central processing unit that receives input from users, processes the data according to system rules, and communicates with the Database for storage and retrieval. The system also generates outputs such as reports, dashboards, and analytics that are used for decision-making and performance monitoring.

The Database serves as the centralized repository of all system data, including sales transactions, inventory records, user accounts, customer information, and promotional data. This centralized structure ensures consistency, accuracy, and real-time availability of information across the system.

User as the Primary External Entity

The User is the main external entity in the Data Flow Diagram. Users interact directly with the system through a web-based or mobile interface. In this system, users are categorized into different roles, each with specific functions and permissions.

1. Staff User Role

The staff user is primarily responsible for daily operational tasks such as:

1. Inputting sales transactions into the system
2. Updating inventory records after each sale
3. Managing customer information
4. Viewing assigned sales reports
5. Processing customer orders

Staff users are the ones who directly interact with the sales processing functions of the system. They ensure that all transactions are recorded accurately and that inventory levels are updated in real time.

For example, when a staff member processes a sale, they input the product details, quantity, and customer information into the system. The system then validates this data, processes the transaction, updates the inventory, and stores the information in the database.

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2. Admin User Role

The administrator (admin) has higher-level access and is responsible for system management and oversight. The admin user performs tasks such as:

1. Adding, editing, and deleting product information
2. Managing promotional tools such as coupons and discounts
3. Managing staff accounts and user permissions
4. Monitoring overall system performance
5. Accessing analytical dashboards and reports

Unlike staff users, admins focus more on system control and decision-making functions rather than daily transaction entry. Their role ensures that the system operates smoothly and that all data remains accurate and organized.

For example, an admin may add a new product to the system, which will then become available for staff to include in sales transactions.

Sales & Management System as the Central Process

The Sales & Management System is the core process in the DFD Level 0 diagram. It acts as the intermediary between the user and the database. All inputs from users are processed through this system, and all outputs are generated from it.
The system performs several key functions:

1. User Authentication and Role Management

When a user logs into the system, the system verifies their credentials and determines their role. This ensures that users only access features that are allowed based on their permissions. For example, a staff user cannot access administrative functions such as deleting products or managing system settings.

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2. Sales Processing

One of the main functions of the system is to handle sales transactions. When a staff user inputs a sale, the system performs the following actions:

1. Validates product availability
2. Calculates total price based on quantity and discounts
3. Updates inventory levels
4. Records transaction details in the database

This ensures that all sales data is accurate and up to date.

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3. Inventory Management

The system continuously updates inventory data based on sales and admin inputs. When a product is sold, the system automatically reduces the stock quantity. If an admin updates product information or adds new stock, the system reflects these changes immediately.
This real-time inventory management helps prevent issues such as overselling or stock inconsistencies.

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4. Customer Data Management

The system also manages customer-related information. Staff users can input customer details such as name, contact information, and purchase history. This data is stored in the database and can be used for future reference or analysis.
Customer data management is important for understanding buying behavior and improving customer service.

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5. Coupon and Promotion Management

Admins can create and manage promotional tools such as coupons, discounts, and special offers. The system applies these promotions automatically during sales processing when conditions are met.

For example, if a customer uses a valid coupon code, the system adjusts the total price accordingly before finalizing the transaction.

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6. Reporting and Analytics Generation

One of the most important outputs of the system is its ability to generate reports and analytics dashboards. The system processes stored data and presents it in a meaningful format for decision-making purposes.
Reports may include:

1. Daily, weekly, or monthly sales summaries
2. Best-selling products
3. Revenue trends
4. Inventory status reports
5. Customer purchase behavior analysis

These reports help business owners and administrators make informed decisions about operations and strategy.

Database as the Central Storage Unit

The Database plays a critical role in the Data Flow Diagram as it serves as the central repository of all system data. It stores and manages all information processed by the system.
The database stores several types of data, including:

1. Sales transactions
2. Product inventory records
3. User accounts and roles
4. Customer information
5. Promotional data such as coupons and discounts
6. System-generated reports and analytics data

The centralized nature of the database ensures that all users access a single, consistent version of the data. This eliminates duplication and reduces the risk of data inconsistencies.

Data Flow Explanation

The flow of data in the system can be explained in a step-by-step process:

Step 1: User Login

The process begins when a user logs into the system. The system verifies the user’s credentials and identifies their role (staff or admin).

Step 2: User Input

After successful login, the user performs actions such as entering sales data, managing products, or updating customer information.

Step 3: Data Processing

The system receives the input and processes it based on business rules. For example, sales data is validated, totals are computed, and inventory is adjusted.

Step 4: Database Interaction

Processed data is stored in the database. The system also retrieves data from the database when needed, such as for generating reports or displaying dashboards.

Step 5: Output Generation

The system generates outputs such as confirmation messages, reports, and analytical dashboards. These outputs are displayed to the user through the interface.

Importance of the DFD Level 0 in the Study

The DFD Level 0 is important because it provides a high-level understanding of the entire system architecture and data flow. It helps researchers and developers visualize how different components of the system interact without focusing on technical implementation details.
It also serves as a foundation for more detailed diagrams such as DFD Level 1 and Level 2, where processes are broken down into smaller and more specific components.
Additionally, the DFD helps ensure that all system requirements are properly addressed. By mapping out data flow, the researchers can identify missing processes, redundant operations, or inefficiencies in the system design.

Integration with System Architecture

The DFD Level 0 is closely related to the system architecture of the Digital Sales Processing System. It reflects the client-server model, where:

1. The User (Client) sends data inputs
2. The System (Server) processes data
3. The Database stores and retrieves information

This alignment ensures that the system design is consistent across both architectural and functional perspectives.

Conclusion of Data Flow Representation

The Data Flow Diagram Level 0 provides a structured representation of how data moves within the Digital Sales Processing System, highlighting the interaction between users, system processes, and the centralized database. It serves as a foundational model that guides the detailed design and development of the system's internal processes and functionalities.

Entity Relationship Diagram (ERD)

The Entity Relationship Diagram (ERD) for the Digital Sales Processing System with Centralized Sales Management and Analytical Reporting represents the logical structure of the database system. It illustrates how different data entities are organized, how they relate to each other, and how they interact within the system to support sales processing, inventory management, customer tracking, and promotional management.

The ERD is a crucial part of the system design because it ensures that the database is properly structured, normalized, and capable of supporting efficient data storage and retrieval. It also helps prevent data redundancy, maintain data integrity, and ensure consistency across all system operations.

In this study, the ERD consists of several key entities, namely User, Sales, Product, Inventory, Customer, and Promo. Each entity represents a real-world object or process within the sales system, and each contains specific attributes that define its properties. The relationships between these entities define how data flows and interacts within the system.

Entities and Their Descriptions

1. User Entity

The User entity represents all individuals who interact with the system. This includes both administrators and staff members. It is one of the most important entities because it controls access to the system and defines user roles and permissions.

Attributes of User Entity:

1. UserID (Primary Key)
2. Name
3. Username
4. Password
5. Role

The UserID serves as the primary key that uniquely identifies each user in the system. The Role attribute is particularly important because it determines whether a user is an admin or a staff member, which directly affects what actions they are allowed to perform.

For example, admins have full control over system management, while staff users are primarily responsible for processing sales transactions.

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2. Sales Entity

The Sales entity represents all sales transactions recorded in the system. It is a central entity in the system because it connects users, products, customers, and promotional data.

Attributes of Sales Entity:

1. SalesID (Primary Key)
2. Date
3. TotalAmount
4. UserID (Foreign Key)

The SalesID uniquely identifies each transaction. The Date attribute records when the transaction occurred, while TotalAmount represents the total value of the sale.

The UserID serves as a foreign key linking the Sales entity to the User entity. This relationship ensures that every sale can be traced back to the user who processed it.

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3. Product Entity

The Product entity stores information about all items available for sale in the system. It plays a key role in both sales processing and inventory management.

Attributes of Product Entity:

1. ProductID (Primary Key)
2. ProductName
3. Price
4. StockLevel

The ProductID uniquely identifies each product. The Price attribute defines the selling price of the product, while StockLevel indicates the current quantity available in inventory.

This entity ensures that product information is centralized and consistently used across sales transactions and inventory updates.

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4. Inventory Entity

The Inventory entity manages stock levels and tracks product quantities over time. It ensures that the system maintains accurate and updated records of available products.

Attributes of Inventory Entity:

1. InventoryID (Primary Key)
2. ProductID (Foreign Key)
3. Quantity
4. LastUpdated

The InventoryID uniquely identifies each inventory record. The ProductID links the inventory record to a specific product. The Quantity attribute indicates how many units are available, while LastUpdated records the most recent update to the inventory.

This entity is essential for preventing stock discrepancies and ensuring real-time inventory accuracy.

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5. Customer Entity

The Customer entity stores information about individuals or clients who make purchases in the system. It helps in tracking purchase history and analyzing customer behavior.

Attributes of Customer Entity:

1. CustomerID (Primary Key)
2. Name
3. ContactInfo
4. PurchaseHistory

The CustomerID uniquely identifies each customer. The ContactInfo stores communication details such as phone number or email address. The PurchaseHistory attribute provides a record of past transactions, which is useful for analytics and customer relationship management.

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6. Promo Entity

The Promo entity represents promotional campaigns, discounts, and special offers applied to sales transactions. It helps in marketing strategies and customer engagement.

Attributes of Promo Entity:

1. PromoID (Primary Key)
2. Discount
3. StartDate
4. EndDate

The PromoID uniquely identifies each promotion. The Discount attribute defines the percentage or value reduction applied. The StartDate and EndDate define the validity period of the promotion.

This entity ensures that promotional rules are properly managed and applied automatically during sales processing.

Relationships Between Entities

The relationships in the ERD define how entities interact with each other within the system. These relationships are essential for maintaining data consistency and enabling meaningful connections between different types of data.

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1. User Processes Sales

This relationship indicates that a User (specifically staff or admin) is responsible for processing sales transactions.

1. One user can process multiple sales.
2. Each sale is processed by only one user.

This is a one-to-many relationship, where one user can be associated with many sales records.

This relationship ensures accountability and traceability of transactions.

For example, if a staff member processes 20 sales transactions, all those transactions will be linked to that specific UserID.

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2. Sales Includes Product

This relationship shows that a Sales transaction includes one or more products.

1. One sales transaction can include multiple products.
2. Each product can appear in multiple sales transactions.

This is a many-to-many relationship, which is typically resolved using a junction table (such as SalesDetails or TransactionItems in actual implementation).

This relationship is important because sales are not limited to a single product; customers often purchase multiple items in one transaction.

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3. Product is Managed in Inventory

This relationship connects the Product entity to the Inventory entity.

1. One product has one inventory record.
2. Inventory tracks stock levels of products.

This is generally a one-to-one or one-to-many relationship depending on system design. In most cases, each product has a corresponding inventory record that tracks its quantity and stock updates.

This relationship ensures that every product sold is reflected accurately in inventory levels.

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4. Customer Makes Sales

This relationship defines the connection between Customer and Sales entities.

1. One customer can make multiple sales.
2. Each sale is associated with one customer.

This is a one-to-many relationship, where customers can have multiple purchase records.

This relationship is important for tracking customer buying behavior and generating purchase history reports.

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5. Admin Manages Staff

This relationship defines administrative control within the system.

1. One admin can manage multiple staff accounts.
2. Each staff member is managed by an admin.

This is also a one-to-many relationship, ensuring proper hierarchy and control within the system.

This relationship supports system security and role-based access control.

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6. Admin Creates Promo

This relationship shows that promotional campaigns are created by administrators.

1. One admin can create multiple promos.
2. Each promo is created by one admin.

This is a one-to-many relationship that ensures accountability in promotional management.

It also helps track which admin created specific discounts or campaigns.

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7. Promo Applies to Sales

This relationship defines how promotions affect transactions.

1. One promo can apply to multiple sales.
2. Each sale may have one or more promos depending on system design.

This is typically a many-to-many relationship, as promotions may be reused across different transactions.
This relationship ensures that discounts are properly applied and reflected in total sales amounts.

Importance of the ERD in the System

The Entity Relationship Diagram is essential in the development of the Digital Sales Processing System because it provides a clear blueprint for database structure. It ensures that all data entities are properly defined and logically connected.
The ERD helps in:

1. Designing a normalized database structure
2. Preventing data redundancy and duplication
3. Ensuring data integrity and consistency
4. Supporting efficient query processing
5. Improving system scalability and maintainability

By clearly defining relationships between entities, the ERD ensures that the system can handle complex operations such as sales tracking, inventory updates, customer management, and promotional applications without data conflicts or inconsistencies.

Role of ERD in System Implementation

During system development, the ERD serves as the foundation for creating actual database tables in Supabase (PostgreSQL). Each entity in the ERD corresponds to a table in the database, while attributes become table columns.
Relationships defined in the ERD are implemented using primary keys and foreign keys, ensuring referential integrity. For example:

1. UserID in Sales table references User table
2. ProductID in Inventory references Product table
3. CustomerID links sales to customer records

This structure ensures that all data relationships are properly maintained within the database system.

Overview of the Use Case Diagram

The Use Case Diagram provides a visual representation of how users interact with the system. It defines the relationship between actors and system functionalities. In this system, all use cases are centered around the Sales Management System, which serves as the main platform for processing sales transactions, managing inventory, and generating reports.
The diagram helps in understanding:

1. What functions are available to each user role
2. How users interact with the system
3. Which features are restricted or authorized based on roles
4. The overall scope of the system's functionality

By clearly defining these interactions, the system ensures that all user requirements are properly addressed during development.

Actors in the System

1. User (Staff/Manager)

The User actor represents employees or staff members who use the system for daily operational tasks. This includes roles such as cashiers, sales staff, and managers who are responsible for handling transactions and monitoring business performance.
The User plays a critical role in ensuring that the system functions properly in a real-world business environment. They are primarily responsible for inputting and managing transactional data, which forms the foundation of the system's analytical reporting features.
The User actor can perform the following functions:

1. Register/Login Account
2. View Dashboard
3. Manage Inventory
4. Process Sales (POS)
5. Generate Reports (depending on access level)

In some system configurations, managers may have additional privileges compared to staff users, such as viewing more detailed reports or accessing broader inventory data.

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2. Administrator

The Administrator actor represents the highest level of authority within the system. The administrator is responsible for managing the entire system, including user accounts, system settings, and overall data control.
The administrator ensures that the system remains secure, organized, and properly maintained. They also have access to all system functionalities, including those restricted from regular users.
The Administrator can perform the following functions:

1. Register/Login Account
2. View Dashboard
3. Manage Inventory

4. Process Sales (POS)
5. Generate Reports
6. Manage Accounts (Admin)

Unlike regular users, administrators have full control over the system and are responsible for maintaining its integrity and security.

Use Case Descriptions

1. Register/Login Account

The Register/Login Account use case is the entry point of the system. It allows users to securely access the system by verifying their identity.

Description:

This use case enables both users and administrators to create an account (if applicable) and log into the system using valid credentials such as username and password. Authentication ensures that only authorized individuals can access the system.

Process Flow:

1. The user opens the system login page
2. The user enters username and password
3. The system validates the credentials
4. If valid, the system grants access based on user role
5. If invalid, the system displays an error message

Importance:

This use case ensures system security and prevents unauthorized access. It also enables role-based access control, ensuring that users only access features relevant to their responsibilities.

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2. View Dashboard

The View Dashboard use case provides users with a centralized overview of system data.

Description:

The dashboard serves as the main interface after login. It displays key performance indicators (KPIs), sales summaries, inventory status, and analytical reports in a visual format.

Features Displayed:

1. Total sales for a specific period
2. Inventory status (low stock, available stock)
3. Top-selling products
4. Recent transactions

5. Revenue charts and graphs

Process Flow:

1. User logs into the system
2. System retrieves relevant data from the database
3. Dashboard displays real-time analytics
4. User interacts with visual reports

Importance:

The dashboard helps users and administrators quickly understand business performance and make informed decisions.

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3. Manage Inventory

The Manage Inventory use case allows users to monitor and update product stock levels.

Description:

This function enables users to add new products, update existing product details, and track inventory levels. It ensures that product availability is always accurate and up to date.

Functions Included:

1. Add new products
2. Update product details
3. Remove discontinued products
4. Monitor stock levels
5. Receive alerts for low stock

Process Flow:

1. User accesses inventory module
2. System displays product list
3. User selects an action (add, edit, delete)
4. System updates database
5. Inventory is refreshed in real time

Importance:

Inventory management ensures that sales operations are not disrupted due to incorrect stock information.

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4. Process Sales (POS)

The Process Sales (Point of Sale) use case is the core function of the system.

Description:

This use case allows users to record sales transactions in real time. It ensures that every sale is properly recorded, calculated, and stored in the centralized database.

Functions Included:

1. Select products for sale
2. Input quantity and customer details
3. Apply discounts or promotions
4. Calculate total amount
5. Save transaction record

Process Flow:

1. User selects products
2. System retrieves product data
3. System calculates total cost
4. Discounts or promos are applied (if any)
5. Transaction is saved in database
6. Inventory is automatically updated

Importance:
This is the most important operational function of the system because it directly handles business revenue generation and transaction recording.

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5. Generate Reports

The Generate Reports use case provides analytical insights based on stored data.

Description:
This function allows users to generate detailed reports based on sales, inventory, and customer data. Reports can be viewed in graphical or tabular formats.

Types of Reports:

1. Daily, weekly, and monthly sales reports
2. Product performance reports
3. Inventory status reports
4. Customer purchase analysis
5. Revenue trend analysis

Process Flow:

1. User selects report type
2. System retrieves relevant data
3. System processes and analyzes data
4. Report is generated and displayed

Importance:

This use case supports decision-making by providing valuable insights into business performance.

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6. Manage Accounts (Admin Only)

The Manage Accounts use case is restricted to administrators.

Description:

This function allows administrators to manage all user accounts within the system. It ensures proper user control and system security.

Functions Included:

1. Add new user accounts
2. Edit user details
3. Assign user roles (staff, manager, admin)
4. Deactivate or delete accounts
5. Reset passwords

Process Flow:

1. Admin accesses account management module
2. System displays list of users
3. Admin selects action (add/edit/delete)
4. System updates user database
5. Changes are applied immediately

Importance:

This ensures that only authorized users have access to the system and helps maintain security and organization.

System Interaction Overview

The Use Case Diagram shows that all system functionalities revolve around two primary actors interacting with a centralized system. The User (Staff/Manager) handles daily operations such as sales processing and inventory updates, while the Administrator manages system control and user accounts.

All actions performed by users are processed through the system and stored in a centralized database. This ensures consistency, accuracy, and real-time data availability across all modules.

The system architecture ensures that:

1. All transactions are recorded centrally
2. All users access real-time data
3. Role-based access controls are enforced
4. Reports reflect accurate and updated information

Importance of the Use Case Diagram in the Study

The Use Case Diagram is important because it clearly defines the functional requirements of the system. It helps developers understand what the system should do from a user perspective before implementation begins.

It also serves as a guide for:

1. System design and development
2. Database structuring
3. Interface design
4. Testing and validation

By clearly defining user interactions, the Use Case Diagram ensures that the system is developed according to actual user needs and business requirements.